Covering the Unit Disk with 100 Equal Disks

A Systematic Account of All Routes, with the Current Upper and Lower Bounds

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Abstract

We study $r_D(100)$, the least radius r such that 100 congruent closed disks of radius r cover the closed unit disk D. This monograph is a systematic, route-by-route account of the whole research programme: every construction that was tried, every relaxation that was closed, and every route that was falsified or left undecided. The currently accepted, machine-replayable statement is

$$\boxed{0.113235 \;<\; r_D(100) \;\le\; 0.1154469548855829}$$

The lower bound is a six-branch exclusion certificate at the rational radius $R = 22647/200000$: $B31$–$B35$ are inherited by radius monotonicity from certificates at $R = 2831/25000 = 0.11324$, while the tail branch $B \ge 36$ is replayed directly at R over 1 841 089 rational boxes. The upper bound is an explicitly replayed 100-centre configuration. Neither statement identifies the exact optimum. The nearby target 0.11324 remains open: with the frozen capped-pair constants the branch $B = 36$ carries an exact rational obstruction point at which the local pair relaxation exceeds the intercept that the $B = 36$ area budget can pay for. We explain why that obstruction cannot be removed by increasing a box budget or raising an intercept, and we record the undecided window (0.113235, 0.11324].

Every quantitative claim below is tied to a machine-readable artifact in `verify/out/`. The document deliberately keeps its failures: a research record that only contains successes cannot tell the next reader which relaxations are too weak.

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Preface: what this document is, and how to read it

This monograph is not a single theorem with a proof. It is the map of a research programme. Covering the unit disk with 100 equal disks is a continuous, non-convex problem whose combinatorial type changes with the radius, and no known method solves it in one step. What exists instead is a set of *routes*: constructions that produce upper bounds, relaxations that produce lower bounds, and a large number of variants that were tried and either closed a sub-case, produced a diagnostic, or were falsified. The purpose of this document is to present those routes systematically, so that a reader can see (i) what the current frontier is, (ii) which inequality carries which part of it, and (iii) exactly where the remaining gap lives.

The one-page state of the art

Item	Value / status
Accepted strict interval	$0.113235 < r_D(100) \le 0.1154469548855829$
Accepted lower bound (exact rational)	$R_\star = 22647/200000 = 0.113235$
Accepted upper bound (exact rational)	$U_\star = 1154469548855829/10^{16}$
Relative gap $(U_\star - R_\star)/U_\star$	approximately 1.92% (display only)
Lower-bound engine	six mutually exclusive boundary-count branches $B31, B32, B33, B34, B35, B{\ge}36$; $B31$–$B35$ inherited from $R = 2831/25000$ by radius monotonicity, $B{\ge}36$ replayed at $R_\star$
Weakest branch	$B \ge 36$, specifically the single value $B = 36$
Target not reached	0.11324; the frozen capped-pair relaxation has an exact obstruction point at $x = y = 9921033/12500000$
Undecided window	(0.113235, 0.11324]; the rational candidates 0.113236 and 0.113237 were started and stopped without producing a certificate
Exact optimum, minimal polynomial	not obtained; `global_optimality_proved = false`
Best universal (configuration-free) lower bound	$2787/25000 = 0.11148$ (long-chord bundle), with the older $554570693/5000000000 = 0.1109141386$ and $1111720153/10^{10} = 0.1111720153$ still valid and dominated

How the parts fit together

- **Part I** fixes the problem, the conventions (closed disks, centres unrestricted, rational decision layer) and the evidence discipline. Read this first: the words *certificate*, *diagnostic*, *analytic* and *inherited* have precise meanings here.
- **Part II** collects the upper-bound routes: how a finite check replaces a continuous coverage statement, the triangular-lattice 97+3 configuration with $\alpha = 1/\sqrt{67}$, the replayed external 100-centre configuration, the cooperative search track, and the local algebraic (KKT) route.
- **Part III** collects the configuration-free analytic lower bounds: the area bound 1/10, the integer-edge-count route with its four geometric lemmas, the sharpened envelope, the long-chord quadratic-loss bound, and the refutation of the perimeter-defect (λ) law.
- **Part IV** is the technical core: the six-branch engine. It explains the same-radius normalization, the counting bridge, and then each branch $B31$–$B35$ and the tail $B \geq 36$, including the capped-pair relaxation and its rational square-domain subdivision.
- **Part V** is the frontier: how the accepted bound moved from 0.11323 to 0.1132301 to 0.113235, why 0.11324 is blocked, and what a genuinely new inequality would have to do.
- **Part VI** is the route catalogue: every route family with its construction, its status and its evidence path, including the falsified and zero-yield ones.
- **Part VII** gives reproduction commands, external dependencies and the boundary of what was absorbed from third-party material.
- **Part VIII** lists the open obligations without softening them.
- **The Appendix** is the evidence ledger: assertion $\leftrightarrow$ artifact, plus the JSON field vocabulary and the exact arithmetic of the $B = 36$ obstruction.

Evidence grades used throughout

Grade	Meaning	Trust
A	structured machine artifact: JSON certificate with status, counts, hashes, exit code	strongest
B	independent replay: a different script or stage rechecks the same sub-assertion	strong
C	discovery material: floating-point search, candidate ranking, failed experiment	guides the next step only
D	inheritance reasoning, e.g. exclusion at a larger radius used at a smaller radius	valid only if recorded explicitly by the combiner

Three further qualifiers appear constantly and must not be confused with grades:

- *analytic* — proved in prose, human-checkable, but not formalized in a proof assistant. Every "universal" statement in this document is analytic plus exact finite arithmetic; none is a kernel-verified theorem.
- `machine_verified_geometry = false` — the finite rational/interval inequalities were machine-checked; the reduction from geometry to those inequalities was not.
- *diagnostic* — the computation deliberately reports a failure, a bottleneck or a zero-yield result and is *not* a certificate that a branch is closed.

Notation

Symbol	Meaning
D	closed unit disk $\{x \in \mathbb{R}^2 : \|x\| \le 1\}$
$B(c, r)$	closed disk of radius r about c
$r_D(n)$	$\inf\{r : D \subseteq \bigcup_{i=1}^{n} B(c_i, r)\}$
R	a fixed rational *candidate* radius used for a contradiction argument
U	a certified upper bound (radius of an explicit covering configuration)
B	number of boundary arcs (equivalently boundary vertices) served by the covering
M	number of interior faces of the normalized common triangulation
N, m	number of distinct centres, resp. number of effectively used disks
k	number of distinct boundary-serving disks
e_i	$m_i + a_i$, the nominal side count of the clipped cell Q_i
Γ	total marked interior angle sum used in the angular ledgers
h	total degree excess of strictly deep vertices
C	number of chords of a core boundary
λ, μ, L, K	penalty coefficient, slope, intercept, marked constant of the tail relaxation
Φ	local pair loss of the capped-pair relaxation

A warning about reading the numbers

Every rational number in this document is quoted as it appears in a certificate or a report. Where a decimal is shown for orientation it is marked as display. Where two stages give slightly different auxiliary constants (for example the normalized centre-norm-squared bound at different radii), both are quoted with their own source rather than silently reconciled. Where a run was started and stopped, it is reported as stopped — not as a failure and not as a proof.

Part I. The problem, the conventions, and the evidence discipline

1.1 Statement of the problem

Definition 1 (Covering radius and the extremal quantity)**.** Let $D = \{x \in \mathbb{R}^2 : \|x\|_2 \le 1\}$. For a centre tuple $C = (c_1, \dots, c_{100}) \in (\mathbb{R}^2)^{100}$ put

$$R_D(C) = \max_{x \in D} \min_{1 \le i \le 100} \|x - c_i\|_2, \qquad r_D(100) = \inf_{C \in (\mathbb{R}^2)^{100}} R_D(C).$$

The covering disks are *closed*; they may overlap and may extend beyond D; the centres are not restricted to D. The task of the programme is to determine $r_D(100)$ together with an explicit minimal polynomial of it, an explicit configuration attaining it, a coverage proof and a global optimality proof, all backed by reproducible computer-assisted material.

The five formal requirements are, in the order in which they constrain the work:

1. **Minimal polynomial and isolation.** List every coefficient of the primitive integer polynomial P of the optimal radius, prove irreducibility over $\mathbb{Q}$, and give rationals $a < b$ with $P(a)P(b) \neq 0$ and exactly one root of P in (a, b).
2. **Explicit configuration.** A definite ordered 100-tuple of centres given by exact algebraic data.

3. **Coverage.** A rigorous proof that every $x \in D$ is within α of some centre; if the continuous statement is reduced to finitely many checks, the reduction itself must be proved sufficient.
4. **Global optimality.** $R_D(C) \geq \alpha$ for *every* C, with no unproved restriction on symmetry, contact structure or combinatorial type, and with degeneracies (coincident centres, redundant disks, collinear centres, tangential contacts) handled explicitly.
5. **Computer-assisted proof.** Complete code, inputs, certificates, environment and actual verification results, with an explanation of why those results imply the mathematical statements.

1.2 Why the problem is hard

1.2.1 Continuous, non-convex, combinatorially unstable

The objective $C \mapsto R_D(C)$ is a maximum of minima of distances: continuous, piecewise smooth, and non-convex. Its minimizers are not characterized by a single smooth system, because the combinatorial type of the induced decomposition changes with r: which centre owns a boundary arc, which triples of centres meet at a Voronoi vertex, and how many arcs the boundary carries are all functions of the configuration. Consequently a numerical optimizer can only ever produce an upper bound. A lower bound must exclude *all* configurations, and the standard way to do that is to pass to a finite combinatorial catalogue of possible decompositions and then exclude each catalogue entry by exact inequalities.

1.2.2 Symmetry is not a necessary condition for optimality

The classical counterexample is $n = 5$: the fully symmetric arrangement gives $r = \varphi - 1 = 0.6180340\ldots$, whereas the true optimum is $0.6093829\ldots$, attained by a configuration with a single mirror axis. This is Connelly's question (must an optimal covering configuration have n-fold rotational symmetry?), answered negatively for $n = 5$ and affirmatively for $n = 3$ by Csikós. For the present programme the practical consequence is blunt: one may *not* restrict to a symmetric family and call the result $r_D(100)$.

1.2.3 Enumeration is not viable at $n = 100$

The mature paradigm for rigorous computer-assisted optimality proofs is *combinatorial classification plus interval Newton / branch-and-bound*. Nurmela and Östergård pushed the rigorous optimality of equal-circle *packings* in a square to $n \leq 27$; their own account records that the initial combinatorial count grows rapidly, that $23 \leq n \leq 26$ cost CPU-months, and that $n = 27$ required a change of algorithm. The present programme's estimate for $n = 100$ is consistent with this: the orbit count of covering configurations is astronomical. Enumeration and classification are therefore used here only as *local* devices inside a branch, never as the global method.

1.2.4 What is known in the literature (snapshot, October 2026)

Unit conversion: the literature usually writes $\rho(n)$ for the largest radius of n unit disks that can cover a disk; then $r_D(n) = 1/\rho(n)$.

n	status	source
1–4, 7	exact closed form, trivial optimality	—
5, 6	exact closed form, optimality proved	K. Bezdek 1979/1983
8, 9, 10	$r_D(n) = 1/(1+2\cos\frac{2\pi}{n-1})$, optimality proved	G. Fejes Tóth 1996/2005
11–20	configurations found; optimality claimed by one group but still listed as unknown by a curated database	Melissen 1997 / Tan 2018; Friedman's `circovcir` page vs. `leapsinbounds`
21–25	configurations found, no optimality proof claim	Tan 2018
26–99	no systematic public table	—
100	no public data beyond lattice-type constructions	—

Three facts from that snapshot shape everything below. First, the asymptotic density statement is rigorous but asymptotic: $\lim_{\varepsilon\to 0^+} \varepsilon^2 N(\varepsilon) = 2\pi/(3\sqrt{3})$, i.e. covering density $\delta = 2\pi/(3\sqrt{3}) = 1.2091996\ldots$ (Kershner 1939; Verblunsky 1949). Converting an asymptotic statement into a finite-n bound requires a boundary correction term, and that correction is precisely where the difficulty sits. Second, the unique successful "symmetry-forcing" paradigm (Fejes Tóth's proof for $n = 8, 9, 10$) works only in the structure "one central disk plus one ring"; $n = 25$ (centre plus three octagonal rings) and $n = 100$ (many rings) have no such theorem. Third, the covering-density inequality of L. Fejes Tóth with a perimeter term, and its packing dual (Oler/Thue–Groemer, $\frac{1}{2\sqrt{3}}A(C) + \frac{1}{4}P(C) + 1 \geq n$), are the literature prototypes of the "cell-by-cell plus boundary correction" route — which Part III shows is *not* strong enough here, and Part IV replaces by a counting-plus-budget engine.

1.3 Conventions and normalizations

1. **Closed everything.** Disks, the unit disk, and all rational intervals are closed. A closed interval $[a, b]$ used as an outer enclosure means the true value lies in $[a, b]$; a closed radius equality is retained rather than discarded.

2. **Decision layer.** Every accepted inequality is decided by integer, `Fraction`, or outward-rounded interval arithmetic. Floating point appears only in the discovery layer: Newton roots, candidate configurations, factor sieving, LLL/PSLQ candidate relations. A discovery result must be re-decided exactly before it can enter a certificate.

3. **Projection.** If $c \notin D$, let $\Pi(c)$ be its projection onto the convex set D. Since Π is non-expansive, $B(\Pi(c), r) \supseteq B(c, r) \cap D$; replacing every centre by its projection preserves both coverage and radius. One may therefore assume $\|c_i\| \leq 1$. This does *not* place the centre inside the inscribed polygon P used in Part III, and one of the geometric lemmas there exists precisely to remove that older assumption.

4. **Deduplication.** If $c_i = c_j$ for $i \neq j$, delete one disk: coverage and radius are unchanged and the centre set shrinks. After deduplication all centres are distinct and every Voronoi cell has positive area.

5. **Non-overwriting outputs.** Certificate scripts refuse to write over an existing JSON target. Re-running a stage therefore requires a fresh output path, which is how the repository distinguishes "this certificate was produced by this run" from "a file with this name exists".

1.4 The evidence discipline

1.4.1 Five layers that must not be conflated

The programme's own integrated report separates five layers of verification, and insists that an earlier layer never substitutes for a later one:

Layer	What it establishes
1. file existence / hash	which bytes this run actually consumed
2. metadata / entry verification	rejection of duplicate nodes, wrong domains, unauthorized entries, and unjustified budget addition
3. exact arithmetic replay	recomputation of the necessary conditions and certificates for a *given* domain
4. analytic / geometric bridge	that every genuine covering configuration falls into that domain, i.e. that the necessary constraints really hold
5. global coverage	that all branches, graphs and labellings are enumerated or enclosed

1.4.2 `PASS` means "passed within the scope field"

A stored `PASS` asserts exactly what the certificate's `scope` field says, no more. Three concrete incidents from this programme make the point:

- **An independent script copied the same error.** In stage 45/46 a boundary-annulus certificate used a wrong inner-face filter, producing an empty pending set; the "globally no-gain" acceptance was voided. The corrected run actually checked 32 boundary-owning faces, 4776 configurations and 152 832 exact annulus comparisons, and added 0 exclusions. The lesson recorded in the state file is that the independent script had duplicated the same filter error, so *exit code 0 is not sufficient evidence that the mathematical coverage domain is right.*
- **A `PASS` on a different radius.** The $B31$ certificate was established at $R = 2831/25000$; it is usable at a smaller radius only through radius monotonicity, and the combiner must record that inheritance explicitly. A `PASS` at one radius is not a `PASS` at another.
- **A `PASS_PREMISES`.** The $B35$ diagnostic certificate returns `status = PASS_PREMISES` with `B35_closed = false`: all the premises were verified and the branch was *not* closed. This state word exists so that such a run can never be read as a closure.

1.4.3 Negative controls are part of the certificate

Almost every certificate script ships destruction tests that must be rejected: a target disk scaled up, an explicit uncovered point for an enlarged-disk variant, a shrunk covering radius, an enlarged covering radius, a wrong metric, a rotated or inflated Voronoi hexagon, a truncated neighbour search, a misattributed neighbour, removed tangency centres, a removed interior centre, a 97-centre configuration submitted as a 100-point claim, a broken edge pairing, a broken face permutation, a support declared inactive, an assertion disabled with `python -O`, and several deliberately corrupted rational payloads. A run whose negative controls are not all rejected is not accepted. Two entry points additionally refuse to run under `python -O` and report *Assertions must remain enabled.*

1.4.4 State words

State	Meaning
`PASS`	the asserted checks passed inside the declared scope
`PASS_PREMISES`	premises verified, closure *not* claimed
`NOT_CLOSED`	the branch or target is open
`EXCLUDED`	the declared domain is empty (a genuine exclusion)
`NOT_EXCLUDED` / `UNDECIDED`	non-empty or unresolved; never counted as excluded
`capped`	propagation hit its round limit without a fixed point; retained, not decided
`diagnostic`	a recorded failure or bottleneck, not a certificate

1.4.5 What must never be inferred

The evidence ledger states the prohibitions explicitly, and they are the reason this document keeps its failures visible:

- a passed local box is not a global lower bound;
- an unblocked obstruction point is not a closed branch;
- an existing configuration is not an optimality proof;
- a numerical "passed" print is not a proof;
- checking finitely many sample points, the boundary, or a numerical image is not a proof that the whole closed unit disk is covered.

Part II. Upper bounds: constructions and their certificates

2.1 What an upper-bound route has to do

An upper bound is a construction: exhibit 100 centres $c_1, \dots, c_{100}$ and a rational U, and prove $\max_{x \in D} \min_i \|x - c_i\| \le U$. The mathematical content is therefore always the same two-step reduction

$$\underbrace{D \subseteq \bigcup_i B(c_i, U)}_{\text{continuous statement}} \impliedby \underbrace{\text{finite list of exact box/leaf checks}}_{\text{machine decision}},$$

and the whole difficulty is in the implication. Three different finite reductions are used in this programme, and they are worth separating because they have different strengths.

2.1.1 Reduction A: quadtree / box subdivision with an exact distance criterion

Cover D by axis-parallel boxes. For a box X and a centre c, an interval evaluation of $\|x - c\|^2$ over X gives $[\ell(X, c), u(X, c)]$. The box is certified if $\min_i \ell(X, c_i) \le U^2$ with the minimum computed exactly; otherwise subdivide. A leaf whose interval cannot be decided is *unresolved* and must be reported, never rounded into "covered". The initial continuous-coverage certificate for the historical configuration is of this type: $R = 153/1250$, 22 804 leaves, depth 16, 0 unresolved, 21/21 checks (`verify/out/coverage_n100.json`). An independent checker re-derives the criterion from

integer corner comparisons and shares no geometric function with the generator; it rejects three deliberately corrupted cases (`verify/out/independent_coverage_n100.json`). A control battery of 12 positive and negative cases passes (`verify/out/coverage_controls_n100.json`). All of these carry `machine_verified_geometry = false`: the reduction from "boxes certified" to "D is covered" is analytic.

2.1.2 Reduction B: lattice-cell reduction (exact, with a finite enumeration)

For a lattice construction one can do much better than a quadtree: the cells tile the plane, so coverage of D reduces to a finite exact check over the finitely many lattice sites whose cells can meet D. This is the method of §2.2, and it produces a certificate whose decisions are integer/Fraction only.

2.1.3 Reduction C: replay of an external configuration

When the configuration is given by an external package, the honest upper-bound route is a byte-faithful replay of that package's own checker plus independent cross-checks. This is the method of §2.3.

2.2 The triangular-lattice 97+3 configuration

2.2.1 The configuration

Work first in *normalized* coordinates where nearest lattice neighbours are at distance 1. A lattice site is a pair $(i, j) \in \mathbb{Z}^2$ with real position

$$x(i,j) = (u, \sqrt{3}\, v), \qquad u = i + \tfrac{j}{2}, \qquad v = \tfrac{j}{2}, \qquad \|x(i,j)\|^2 = i^2 + ij + j^2.$$

The hexagonal Voronoi cell of a site is the translate of the hexagon H with vertices $(0, \pm\frac{1}{3})$, $(\pm\frac{1}{2}, \pm\frac{1}{6})$ in (u, v) coordinates, whose circumradius is $1/\sqrt{3}$.

Construction 1 (The 97+3 configuration)**.** Take all sites with $i^2 + ij + j^2 < 28$ (there are 97 of them), plus the three norm-28 sites

$$(-6, 4), \quad (-2, -4), \quad (2, 4).$$

With the scaling

$$a = \sqrt{3/67}, \qquad R = 1/a = \sqrt{67/3}, \qquad \alpha = 1/\sqrt{67},$$

the 100 closed disks of radius α about the scaled centres $a\, x(i,j)$ cover D, and α is *exactly* the covering radius of this centre configuration.

Proposition 1 (Exact covering radius of the configuration)**.** $\alpha = 1/\sqrt{67}$ *is the least radius at which the* 100 *centres of Construction 1 cover* D*.*

Argument. The upper half is the cell check described below. For the lower half, consider the interior lattice hole $q = (u, v) = (\frac{1}{2}, \frac{1}{6})$, i.e. the centre of a hexagon surrounded by six sites. Its normalized norm is $\|q\|^2 = u^2 + 3v^2 = \frac{1}{4} + \frac{1}{12} = \frac{1}{3}$, and its distance to the nearest lattice site is the hexagon circumradius $1/\sqrt{3}$. After scaling by a, the nearest retained centre is at distance $a/\sqrt{3} = 1/\sqrt{67} = \alpha$, while $\|aq\| = a/\sqrt{3} = \alpha < 1$, so aq is a point of D at distance exactly α from every centre. Hence no smaller radius covers D. □

Remark 1. The three norm-28 sites are *redundant* for coverage: the 97-site configuration already covers D with the same radius, and the certificate records the construction order (97 sites, then three extra sites). Redundancy does not prove that a smaller radius is achievable with 100 disks; it only shows that this particular configuration does not use its budget.

2.2.2 The exact cell argument

The certificate `verify/out/lattice97_upperbound_n100.json` (schema `n100-lattice97-upperbound-v1`) reduces continuous coverage to a finite exact check by four steps, each of which is an integer or `Fraction` decision:

1. **Cell containment.** Every point of the Voronoi cell of a site p is within $1/\sqrt{3}$ of p (circumradius of H).
2. **Finite site list.** A site whose cell meets the closed disk of radius $R = \sqrt{67/3}$ satisfies $\|p\|^2 < 29$, hence $\|p\|^2 \le 28$, hence $|i|, |j| \le 7$. The enumeration box used is $|i|, |j| \le 24$, generously larger than the sufficient box 7.
3. **Exact minimum at a unique vertex.** For every site $p \notin C$ with $\|p\|^2 \le 28$, the minimum of $\|x\|^2$ over the cell is exactly $R^2 = 67/3$, attained at a unique vertex v_p; this is certified by the supporting-plane inequality $\langle v_p,\, w - v_p\rangle \ge 0$ over all six vertices w of the cell.
4. **Exterior shell.** Sites with $\|p\|^2 \ge 29$ have cell distance $\ge \sqrt{29} - 1/\sqrt{3} > R$, certified by the pure integer inequality $3(29-22)^2 > 4 \cdot 29$, i.e. $7 > 2\sqrt{29/3}$.

2.2.3 Algebraic data of the candidate radius

$$\alpha = \frac{1}{\sqrt{67}}, \qquad P(t) = 67t^2 - 1 \;\; \text{(coefficients } [67, 0, -1]),$$

P is primitive with positive leading coefficient; 67 is not a square, so P is irreducible in $\mathbb{Q}[t]$; and

$$\frac{1221694443}{10^{10}} < \alpha < \frac{1221694444}{10^{10}}$$

is a rational isolating interval. The endpoint signs are decided by integer square comparison, and P is strictly increasing on the positive half-line ($P'(t) = 134t > 0$), so the interval contains exactly one root. This is the minimal polynomial of *this configuration's* covering radius — it is *not* claimed to be the minimal polynomial of $r_D(100)$, because the lower bounds of Part III/IV do not meet α.

2.2.4 Negative controls in the certificate

The lattice certificate's thirteen negative controls must all be rejected, and are: a scaled-up target disk (precondition only, explicitly *not* an uncovered witness for the delivered claim); an explicit uncovered point for the enlarged-disk variant; a shrunk covering radius; an enlarged covering radius; a wrong metric; a wrong/rotated hexagon; an inflated hexagon; a truncated neighbour search; a misattributed neighbour; two removed tangency centres; the 97-centre configuration submitted as a "100-point" claim; a single shell centre removed (coverage still holds, so only the cardinality is rejected); and a removed interior centre. An independent implementation (`verify/out/independent_lattice97_upperbound_n100.json`, `all_passed = true`, `coverage_confirmed = true`) reproduces the claim with a different code path.

2.3 The external 100-centre configuration

The best currently accepted upper bound comes from an external public package at the frozen commit

`67ebf1c5776f72a7b7786a8655eed5463429ae55`,

whose bytes were replayed locally without modification. It provides a 100-centre configuration and the exact rational bound

$$U_\star = \frac{1154469548855829}{10^{16}} = 0.1154469548855829.$$

Artifact	What it establishes
`verify/out/external-disk100-bounds-67ebf1c.json`	the complete interval evidence: exit 0; all six lower-bound branches recomputed `PASS`
`verify/out/external-disk100-margin-67ebf1c.json`	an independent upper-bound margin check; exit 0
`verify/out/external-disk100-local-quadtree/replay.json`	this project's own quadtree cross-check at the slightly wider $R = 0.115447$: main and independent `PASS`, 0 unresolved

Two boundary remarks belong here. First, the same package also supplied the *six-branch skeleton* that Part IV uses; the absorption report fixes the roles: the third-party material is a replayable input and an external reference, not an authority automatically inherited by this project. Second, the replay is a byte-faithful run of the package's own checkers plus independent cross-checks; it is *not* a claim that the package's analytic geometry has been formalized.

2.4 The cooperative-search tightening track

Before the external configuration was absorbed, this project tightened its own configuration through a cooperative (free-point plus annealing) search. Each stage froze its output read-only and certified a rational upper bound with an independent quadtree replay:

Stage	floating-point discovery radius (not a decision)	certified rational upper bound	evidence
exact lattice	0.1221694443563052	$1/\sqrt{67}$ (algebraic)	`n100-algebraic-bounds-bundle.json`
cooperative, round 1	0.1200703574813639	$18761/156250 = 0.1200704$	`n100-cooperative-anchor-tight/replay.json`
refine	0.1198908685151380	$119891/1000000 = 0.119891$	`n100-cooperative-refine-tight/replay.json`
polish	0.1198764764337424	$119877/1000000 = 0.119877$	`n100-cooperative-polish-tight/replay.json`

The replay certificates report `all_passed = true`, `independent.verified = true`, 35 584 leaves, maximum depth 23, 0 unresolved for the polish stage; the decimal coordinates of the frozen candidate are treated as *exact rationals*, not replaced by fitted lattice or algebraic coordinates. Earlier stages of the same track certified $153/1250 = 0.1224$ (stage 1) and $1223433/10000000 = 0.1223433$ (stage 2, server search with six candidates, 916 objective evaluations in 15.746 s, all six stopping after 150 consecutive non-improvements).

2.4.1 Why the cooperative track stopped improving

The algebraic diagnostics are as important as the bounds. At every stage there was exactly one tight hole (threshold 10^{-9}), with several near-tight holes participating in a finite band model. The polish run still recorded 115 strict improvements when it hit its 4×30-round cap, so the end of the descent had not been reached: the local system has *no* KKT or neighbourhood-optimality certificate. A stationarity diagnostic ran in 1.3 s and reported: a 10^{-6} near-active band with 123 contacts, but only 2 contacts at 10^{-7} and 1 at 10^{-9}; 114 interior gradient rows of floating-point rank 114 with nullspace dimension 86; and no non-negative KKT multiplier LP found at the tested tolerances. A contact-balance/SQP unit returned `NO_IMPROVEMENT`: in 60 s it lowered the radius by about 1.25×10^{-9}, below the acceptance threshold, so no new candidate was saved and no upper bound was updated. Smooth-max directions mostly increased the true radius and were rejected by the true evaluator. All of this is finite floating-point diagnosis: it neither proves that no other equilibrium branch exists nor that the candidate is "non-algebraizable".

2.5 The local algebraic route: an isolated KKT root

Independently of the configuration search, the local algebra of the absorbed candidate was replayed. The isolated algebraic KKT root has

Quantity	Value
variables	932
basic variables	493
constraints	439 = 406 distance + 32 boundary + 1 gauge
monomials	8 348, degree at most 2

The certificate `verify/out/external-disk100-strict-local-67ebf1c.json` records a fresh root, a fresh coverage check, and 8/8 destruction tests rejected; a Krawczyk-style box isolates the root and the rotation quotient is handled. The stress-transfer analysis on the same root box (radius 10^{-58}) found 406 distance multipliers strictly positive inside the box, 32 boundary multipliers strictly negative, and a 246-node connected contact graph whose weighted Laplacian block over the 214 free nodes, anchored at the 32 boundary contacts, has an exact positive definite lower bound. However, the attempted cover-19-style global Dirichlet energy lower bound failed: the Loewner matrix built from boundary-angle cosine intervals and rod-graph hop distances still produced two negative LDL pivots, so the transfer result is recorded as `H0_PSD = false`. That failure is a route diagnostic, not a counterexample to the local second-order condition and not a statement about other topologies.

2.6 What the upper bounds do *not* give

- They do not identify $r_D(100)$: the accepted interval has width 0.0022 in absolute terms.
- They do not prove that the lattice configuration, the cooperative candidate, or the external configuration is optimal or locally optimal (except for the one isolated KKT root, where *strict local* minimality of the fixed contact graph is certified — a local statement about one graph).
- They do not yield the minimal polynomial of $r_D(100)$. What exists is the minimal polynomial $67t^2 - 1$ of the covering radius of one explicit configuration, which is a different object.
- The three redundant lattice disks show that this configuration does not saturate its budget; that is a hint, not a proof, that a smaller radius might be reachable with 100 disks.

Part III. Universal analytic lower bounds (configuration-free routes)

3.1 Two kinds of lower bound

A lower bound on $r_D(100)$ can be obtained in two essentially different ways.

1. **Universal (configuration-free) bounds.** A single analytic inequality, valid for every covering by 100 disks of any radius r in a range, contradicts coverage below some threshold L. No combinatorics of the specific configuration enters; the price is that such bounds are comparatively weak.
2. **Candidate-radius contradiction.** Fix a rational R and assume a covering at radius R exists; then derive a finite, exhaustive combinatorial classification of all possible configurations and exclude each class by exact inequalities. This is the engine of Part IV, and it is what currently carries the bound 0.113235.

Part III is the first kind. It matters for three reasons: it gives a floor that needs no combinatorial catalogue, it calibrates how much of the gap each ingredient can possibly close, and it produced two instructive refutations (the perimeter-defect law, and simple global energies) that redirected the programme towards Part IV.

3.2 The area bound and its immediate sharpening

The crudest bound is exact and trivial:

$$\pi = \operatorname{area}(D) \le \sum_{i=1}^{100} \operatorname{area}\big(B(c_i, r) \cap D\big) \le 100\pi r^2 \quad \Longrightarrow \quad r \ge \frac{1}{10}.$$

The asymptotic density statement $\delta = 2\pi/(3\sqrt{3}) = 1.2091996\ldots$ replaces 1 by δ in the limit $\varepsilon \to 0$, giving $\sqrt{\delta/100} = 0.109964$ for a hypothetical large-n configuration with negligible boundary. Both numbers are far below the truth, and the gap between them and the accepted upper bound $0.11544\ldots$ is the "boundary correction" that every serious route must earn.

3.3 Route: integer edge counting, the repaired Euler route

This is the most completely written analytic route in the programme. It proves the exact statement

$$r_D(100) \;\ge\; L_1 := \frac{554570693}{5000000000} = 0.1109141386,$$

and its interest is that it repairs, rather than discards, the historical "Euler + boundary cell" chain whose audit had downgraded it to *unproved* with ten explicit obligations O1–O10.

3.3.1 Setup: Voronoi cells, arcs, and the inscribed polygon

Assume 100 disks $B(c_i, r)$ with $r < 1$ cover D (if $r \geq 1$ the conclusion is trivial), and apply the projection and deduplication normalizations of Part I so that $\|c_i\| \leq 1$ and the c_i are distinct. Define the closed convex Voronoi cells

$$V_i = \{x \in D : \|x - c_i\| \leq \|x - c_j\| \ \ \forall j\}.$$

Three facts are used constantly:

- $\bigcup_i V_i = D$ and $\operatorname{int} V_i \cap \operatorname{int} V_j = \emptyset$ for $i \neq j$;
- $V_i \subseteq B(c_i, r)$: for $x \in V_i$, $\|x - c_i\| = \min_j \|x - c_j\| \leq r$ by coverage;
- after projection and deduplication every V_i has positive area, so there is no "empty cell" bookkeeping problem.

Let the *arcs* $A_1, \ldots, A_{B_{\text{arc}}}$ be the maximal positive-measure connected segments of ∂D on which the ownership (smallest index attaining the minimum distance) is a.e. constant; isolated zero-measure ties do not cut an arc. Let the *boundary vertices* be the endpoints of the arcs, and let

$$P := \operatorname{conv}(\text{all boundary vertices})$$

be the inscribed polygon.

Lemma 1 (Arc length bound; and an erratum)**.** *For $r < 1$ every arc has angular measure $\leq 2 \arcsin r < \pi$. Consequently $B_{\text{arc}} \geq \pi / \arcsin r$, and no arc wraps around the whole circle.*

Proof. If an arc had measure $\psi \geq \pi$ it would contain a sub-arc of measure π whose two endpoints are antipodal, hence at distance 2; both lie in $V_i \subseteq B(c_i, r)$, so $2r \geq 2$, i.e. $r \geq 1$, contradiction. For $\psi < \pi$ the chord formula applies: with chord length $\ell \leq 2r$, $\psi = 2 \arcsin(\ell/2) \leq 2 \arcsin r < \pi$. Summing over arcs gives $2\pi \leq B_{\text{arc}} \cdot 2 \arcsin r$. □

Remark 2 (A corrected constant, kept on the record). An early draft used $4 \arcsin(r/2)$ as the upper bound for the angular measure of $\partial D \cap B(c_i, r)$. That is wrong: it is only the value at $\|c_i\| = 1$, which is *smaller* than the true maximum. With $Q(\rho) = (1 + \rho^2 - r^2)/(2\rho)$, the measure is $2 \arccos Q(\rho)$, and Q is minimised at $\rho_* = \sqrt{1 - r^2}$, giving the maximum $2 \arccos \sqrt{1 - r^2} = 2 \arcsin r$. The wrong constant cannot imply $< \pi$ for r near 1 and is therefore unusable; the erratum is retained in the source.

Lemma 2 (Lemma A: area of a convex polygon inside a disk, centre unrestricted)**.** *Let $K \subset \mathbb{R}^2$ be convex and compact with $K \subseteq B(c, r)$, where c need not belong to K, and let k be the number of extreme points of K. Then*

$$\operatorname{area}(K) \leq r^2 f(k), \qquad f(k) = \tfrac{k}{2} \sin \tfrac{2\pi}{k} \ \ (k \geq 3), \qquad f(0) = f(1) = f(2) := 0.$$

Sketch of the five steps. Let $B(o, R)$ be the minimal enclosing circle of K, so $R \leq r$. The contact set $S = K \cap \partial B(o, R)$ consists of vertices and is non-empty. A finite separation argument shows $o \in \operatorname{conv}(S)$: if not, translating o slightly towards $\operatorname{conv}(S)$ strictly decreases the distance to every contact vertex and, by finiteness, to every other vertex as well, contradicting minimality. Hence $o \in K$. Since $o \in K$, the triangles (o, v_j, v_{j+1}) over the extreme points $v_1, \ldots, v_m$ exactly partition K (with the boundary and vertex positions of o treated separately; the case "o is an extreme point" is excluded because a non-negative combination $\sum_s \lambda_s u_s = 0$ inside a convex cone of opening

$< \pi$ is impossible). Therefore $\text{area}(K) \le \frac{R^2}{2} \sum_j \sin\alpha_j$ with $\alpha_j \in [0,\pi]$ and $\sum_j \alpha_j = 2\pi$. Jensen's inequality for the concave sine gives $\sum_j \sin\alpha_j \le m \sin(2\pi/m)$, and $M \mapsto M\sin(2\pi/M)$ is increasing for $M \ge 3$ because $x \mapsto \sin x/x$ is strictly decreasing on $(0,\pi)$. Finally $m \le k$ and f is increasing on the integers ≥ 3, while $m \le 2$ gives area 0. □

Remark 3. Lemma 2 removes the historical hypothesis $c_i \in V_i \cap P$. The sector decomposition is taken from the minimal enclosing circle's centre o, and $o \in K$ is *proved*, not assumed. The bound is optimal: equality holds for a regular k-gon inscribed in the circle of radius r.

Lemma 3 (Lemma B: the caps belong to their own cell; $Q_i = V_i \cap P$ exactly)**.** *For each arc A let $C(A) = \text{conv}(A)$ be the circular segment cut off by its chord, and let H_A^{in} be the closed half-plane bounded by the chord that contains the origin.*

1. *If the arc A belongs to cell V_i (i.e. $A \subseteq V_i$), then $C(A) \subseteq V_i$.*
2. *For $i \ne j$ and an arc A of cell j, $V_i \subseteq H_A^{\text{in}}$.*
3. *Consequently, with $Q_i := V_i \cap P$,*

$$Q_i = V_i \cap \bigcap_{A \text{ of cell } i} H_A^{\text{in}} \quad \textit{(exact identity)},$$

$$\bigcup_i Q_i = P, \quad \text{int}\, Q_i \textit{ pairwise disjoint}, \quad Q_i \subseteq B(c_i, r).$$

4. *$\#\text{sides}(Q_i) \le m_i + a_i =: e_i$, where m_i is the number of maximal straight edges of V_i and a_i the number of arcs owned by cell i; and $\sum_i a_i = B_{\text{arc}}$.*

Sketch. (1) $C(A) = \text{conv}(A)$ by a two-line convexity argument, and V_i is convex. (2) If $x \in V_i \cap \text{int}\, C(A)$ then, since $\text{int}\, C(A) \subseteq \text{int}\, V_j$ and interiors of distinct cells are disjoint, we get a contradiction; the boundary case follows by a limiting argument along a segment from an interior point of V_i. (3) Intersecting all half-planes and deleting the redundant ones by (2). (4) Each owned chord half-plane adds at most one straight side and cannot split an existing straight side; degenerate Q_i (point or segment) has area 0. □

Remark 4 (The risk point, answered). The natural worry is that $V_i \cap P$ might acquire edges coming from *other* cells' chords. Lemma 3(2) shows it cannot: every cap is contained in the convex hull of its own cell, so $V_i \cap P$ differs from V_i exactly by its own caps. This holds for any number of arcs, any degeneracies and any r.

Lemma 4 (Lemma C: lower bound for the area of P)**.** *If $g_1, \dots, g_B \ge 0$, $\sum_k g_k = 2\pi$ and $g_k \le \gamma \in (0,\pi]$, then*

$$A := \tfrac{1}{2} \sum_k \sin g_k \;\ge\; \pi \frac{\sin\gamma}{\gamma} \;\ge\; \pi\Big(1 - \frac{\gamma^2}{6}\Big) \qquad (\gamma < 1).$$

Proof. $\sin g_k = g_k\, s(g_k) \ge g_k\, s(\gamma)$ because $s(t) = \sin t/t$ decreases on $(0,\pi]$; sum. The second inequality is the alternating-series bound for $\sin\gamma/\gamma$. □

Lemma 5 (Lemma D: planar-graph edge accounting, no general position)**.** *Build the boundary Voronoi complex: vertices are the interior Voronoi vertices (where ≥ 3 cells meet in a positive wedge) together with the B_{arc} boundary vertices; edges are the bisector segments,* cut at every graph

vertex*, together with the* B_{arc} *arc edges; faces are the* N *cells plus the outer face. Then every vertex has degree* ≥ 3*, and*

$$\sum_{i=1}^{N} e_i \;\leq\; 6N - 6 - B_{\mathrm{arc}}.$$

Sketch. Degrees: at an interior vertex the cells meeting there fill a full 2π of wedges separated by positive-length bisector rays, so at least three; at a boundary vertex the arc ends from the two sides plus at least one bisector ray entering $\operatorname{int} D$ give degree ≥ 3 (the bisector meets ∂D transversally because the sign of $\varphi(\theta) = \|u(\theta) - c_k\|^2 - \|u(\theta) - c_i\|^2$ changes there, so $\varphi' \neq 0$). Euler without assuming connectivity, with $K \geq 1$ components, reads $E_s = V_{\mathrm{int}} + N - K \leq V_{\mathrm{int}} + N - 1$. Double counting gives $\sum_i m_i \leq 2E_s$, and $\sum_i a_i = B_{\mathrm{arc}}$. Hence $2E_s \geq 3V_{\mathrm{int}} + B_{\mathrm{arc}}$ yields $V_{\mathrm{int}} \leq 2N - 2K - B_{\mathrm{arc}}$, $2E_s \leq 6N - 6K - 2B_{\mathrm{arc}} \leq 6N - 6 - 2B_{\mathrm{arc}}$, and finally $\sum_i e_i \leq 6N - 6 - B_{\mathrm{arc}}$. □

Remark 5 (Exact counterexamples showing the hypotheses cannot be dropped). (i) $n = 1$, $c_1 = 0$, $r = 1$: the single arc wraps the circle, the two endpoints coincide, the identity #boundary vertices $= B_{\mathrm{arc}}$ fails, and $\sum m_i = 0$ violates $6N - 6 - 2B_{\mathrm{arc}} = -2 < 0$. This degeneration is impossible for $r < 1$ by Lemma 1. (ii) The three-centre configuration $c_1 = 0$, $c_2 = (a, 0)$, $c_3 = (-a, 0)$ with $0 < a < 2$ has $V_1 = D \cap \{|x_1| \leq a/2\}$ owning *two* arcs, V_2, V_3 being cap-only cells with $e = 2$, so $B_{\mathrm{arc}} = 4$ while the number of boundary-touching cells is 3; and $\sum_i e_i = 8 = 6N - 6 - B_{\mathrm{arc}}$ is tight. Reading B as the number of boundary cells loses one unit of strength; assuming "one arc per boundary cell" would record e_1 as 3 instead of 4 and is simply false.

3.3.2 The seven-step aggregation

Assume a covering at radius $r \leq L_1$ exists, with $r < 1$.

(1) *Boundary structure.* By Lemma 1, $B_{\mathrm{arc}} \geq \pi / \arcsin r$; since $\pi / \arcsin L_1 = 28.2663 > 28$ and B_{arc} is an integer, $B_{\mathrm{arc}} \geq 29$. Also $B_{\mathrm{arc}} \geq 3$, so P is a genuine polygon.

(2) *Partition and area identity.* By Lemma 3, $\operatorname{area}(P) = \sum_i \operatorname{area}(Q_i)$ with $N \leq 100$, each $Q_i \subseteq B(c_i, r)$ convex and $\#\mathrm{sides}(Q_i) \leq e_i$.

(3) *Per-cell area bound.* By Lemma 2 and monotonicity of f on integers ≥ 3, $\operatorname{area}(Q_i) \leq r^2 f(e_i)$; cells with $k_i \leq 2$ contribute $0 = r^2 f(e_i)$. Hence $\operatorname{area}(P) \leq r^2 \sum_i f(e_i)$.

(4) *Edge accounting.* Lemma 5 gives $\sum_i e_i \leq 6N - 6 - B_{\mathrm{arc}} \leq 600 - 6 - 29 = 565$.

(5) *Affine secant accounting.* Put

$$a := f(6) - f(5) = 0.2204349206\ldots > 0, \qquad b := 6f(5) - 5f(6) = 1.2754666876\ldots > 0,$$

and let $\varphi(e) = ae + b$ be the secant through $(5, f(5))$ and $(6, f(6))$. For integers $e \geq 3$ one has $f(e) \leq \varphi(e)$: for $e \geq 6$ this follows from concavity of f on $[3, \infty)$ ($f''(M) = -2\pi^2 \sin(2\pi/M)/M^3 < 0$), and $e = 3, 4$ are checked exactly,

$$\varphi(3) - f(3) = \tfrac{15}{4}\big(2\sin 72^\circ - \sqrt{3}\big) > 0, \qquad \varphi(4) - f(4) = 5\sin 72^\circ - \tfrac{3}{2}\sqrt{3} - 2 = 0.1568\ldots > 0.$$

Cells with $e \leq 2$ are deleted (they contribute 0). Hence

$$\sum_i f(e_i) \leq a \sum_i e_i + bN \leq 565a + 100b = 65f(6) + 35f(5),$$

the last step being the algebraic identity $565(f_6-f_5)+100(6f_5-5f_6) = 65f_6+35f_5$. Numerically $565a + 100b = 252.0923989138\ldots$ This device replaces "integer concavity plus 92 separate sign checks" by two finite inequalities and, crucially, is *strict*: the continuous relaxation would give $100f(5.65) = 253.2941376702\ldots$, so the discrete bookkeeping buys 1.2017387564 ($\approx 0.48\%$).

(6) *Area lower bound.* By Lemmas 4 and 1 with $\gamma = 2\arcsin r$, $\operatorname{area}(P) \geq \pi(1-\gamma^2/6)$.

(7) *Closing.* Combining (3)–(5) with (6),

$$\pi\big(1-\tfrac{1}{6}(2\arcsin r)^2\big) \;\leq\; r^2\big(65f(6)+35f(5)\big) \qquad \text{for all } r \leq L_1.$$

The left side decreases and the right side increases in r, so it suffices to refute the inequality at $r = L_1$. Exactly,

$$65f(6)+35f(5) = \frac{195\sqrt{3}}{2} + \frac{175}{8}\sqrt{10+2\sqrt{5}},$$

$$\pi\big(1-\tfrac{1}{6}(2\arcsin L_1)^2\big) = 3.1157211653\ldots \;>\; L_1^2 \cdot 252.0923989138\ldots = 3.1012271140\ldots,$$

a margin of 0.0144940513 (0.467% in relative terms). The final step is a finite algebraic/rational inequality, closed by the arithmetic agent with 15 exact checks using rational enclosures of π and $\arcsin L_1$. Therefore no covering at radius $\leq L_1$ exists:

$$r_D(100) \;\geq\; L_1 = \frac{554570693}{5000000000} = 0.1109141386.$$

Remark 6 (What the repair consisted of). The historical chain used the exact extremum of $\operatorname{area}(P)_{\min}(r, B)$ and a continuous fixed point whose root lies 7.6×10^{-12} below L_1. The repaired route deliberately uses the *coarser* lower bound $\pi(1-\gamma^2/6)$, giving up about 0.46% of accuracy, and in exchange eliminates every question about the value of B and the distribution of P's vertices — which is what makes the final inequality strict and finite. The 0.467% net margin comes entirely from the strict discrete secant bookkeeping of step (5).

Historical obligation	Status after the repair
O1 degenerate degree-≥ 4 vertices	absorbed: only degree ≥ 3 plus the inequality $\sum_i m_i \leq 2E_s$ is needed
O2 degenerate boundary vertex changes b	B is defined as the arc count; each arc $<\pi$ for $r < 1$
O3 well-definedness of b, multi-arc cells	solved: $e_i = m_i + a_i$, with an exact counterexample
O4 sector decomposition needs $c_i \in V_i \cap P$	removed: Lemma 2 uses the minimal enclosing circle centre
O4′ cap term bound	removed: caps are owned by their cell (Lemma 3)
O5 centres outside D	projection normalization; $c_i \in P$ no longer needed
O6 coincident centres	deduplication; every cell has positive area
O7 collinear triples / parallel bisectors	solved: only Euler, degree ≥ 3, and the $\leq$ direction
O8 redundant disks / empty cells	no empty cells after projection and deduplication
O9 boundary degeneracies (tangency, single-point arcs)	zero-measure contacts create no arc; a full-circle arc is excluded
O10 fixed point / monotonicity upgrade	removed: replaced by one finite inequality at $r = L_1$

3.3.3 What remains open in this route

The geometric part depends on one finite inequality whose exact arithmetic is delegated; and the step "cells meeting at a boundary vertex fill a half-neighbourhood" is a geometric argument without a machine certificate. The route does not assume general position, so no perturbation or limit argument is needed; if one is used anyway, the correct form is to take $\varepsilon < (L - r)/2$ and enlarge the radius to $r + \varepsilon < L$ *with* $r + \varepsilon \le L$ re-established by the choice of ε — the mistake to avoid is to perturb past L and keep using $r + \varepsilon \le L$.

3.4 Route: the sharpened envelope

The same skeleton — projection, convex Voronoi cells, multi-arc Euler accounting, integer edge counting — was then pushed with re-certified parameters

$$\arcsin L < T, \qquad 28T < \pi,$$

which force $B \ge 29$ and $\sum_i e_i \le 565$, while the area upper-bound coefficient keeps its exact form $G = 65f_6 + 35f_5$ and the area lower bound keeps the sharper $\pi \sin(2 \arcsin r)/(2 \arcsin r)$ (instead of the truncated series). An alternating rational polynomial lower bound then excludes every $0 < r \le L$ at once. The result is

$$r_D(100) \;\ge\; L_2 := \frac{1111720153}{10^{10}} = 0.1111720153,$$

an improvement of exactly $2578767/10^{10} = 0.0002578767$ over the previous value, with a positive margin of about $2.5235222738557335 \times 10^{-9}$ stored as an exact fraction in the certificate. The main arithmetic performs 16 exact checks; the independent arithmetic uses arctan and square-root envelopes, shares no series implementation with the main one, and passes 40 checks plus 9 negative controls. The ideal envelope of this route has a unique zero isolated in $(1111720153/10^{10}, 1111720154/10^{10})$ — a statement about the *method's limit*, not an interval for $r_D(100)$. Higher precision cannot close the gap to the upper bound along this route. Sources: `proofs/n100-sharp-envelope-lowerbound.md`, `docs/review-sharp-envelope-n100.md`, `verify/out/n100-algebraic-bounds-bundle.json`.

3.5 Route: long-chord quadratic loss

A different universal bound comes from a quadratic deficit associated with long chords of the boundary arcs. After three rounds of independent review the geometric draft was frozen at SHA-256 `658863d2a36118be915fda66f10596f05073aadd51738da55bcf35f967d8cf73`, and the main and independent rational-arithmetic checkers replay through `verify/out/n100-longchord-bundle-frozen.json`: unified exit 0, sub-exit codes $[0, 0]$, `all_passed = true`, `universal_lower_bound_accepted = true`, all 566 arc counts enumerated with positive margin, worst case $B = 42$, and 6/6 negative controls rejected. This yields

$$r_D(100) \;\ge\; L_3 := \frac{2787}{25000} = 0.11148,$$

which dominates both L_1 and L_2 and is the strongest configuration-free bound currently accepted. The geometry is analytic proof plus review (`machine_verified_geometry = false`); it is not a formalized proof.

Remark 7 (Layering inside the frozen bundle, kept visible). The same bundle is more conservative in its own sub-blocks than at the top level, and the distinction matters. The top-level fields report `universal_lower_bound_accepted = true` and the primary and independent arithmetic rows agree on a minimum margin of about 4.4799×10^{-4} over 566 rows. Inside the geometry sub-block, however, `all_geometric_premises_closed = true` while `combined_universal_lower_bound_accepted = false`, and the arithmetic is marked `reported_exit_0_by_parent_agent_not_recomputed_and_not_accepted_here`: for a numeric threshold claim the chain additionally needs the premises (P1) and (P2) certified by exact rational arithmetic *and* an independent check, and that extra step lives in the separate arithmetic certificates rather than inside the frozen geometry block. Quoting the top-level acceptance without this qualifier would overstate what the geometry block itself certifies.

3.6 Refuted universal routes

3.6.1 The perimeter-defect (λ) law is impossible

An early programme plan proposed to prove a law of the form

$$nr^2 \;\geq\; \delta\,(1+\lambda r), \qquad \delta = \frac{2\pi}{3\sqrt{3}} = 1.2091996\ldots,$$

and to tune λ so that $n = 100$ is pinned to within 0.4%. That plan is dead, and the reason is a two-line computation. If such a law holds for some n, then at that n's proved optimal value r_n it forces

$$\lambda \;\leq\; \frac{n\,r_n^2/\delta - 1}{r_n},$$

and the proved values give 0.697 ($n = 8$), **0.669** ($n = 9$), 0.734 ($n = 10$), 0.752 ($n = 19$), 0.763 ($n = 25$). So a global law with $\lambda \approx 0.87$ cannot hold — $n = 9$ alone refutes it — and the largest admissible constant, $\lambda^* = 0.6688$, gives at most $r_D(100) \geq 0.114081$, i.e. 6.75% below the best known configuration. Worse, the area–perimeter implementation that was supposed to supply the perimeter layer is never binding cell-by-cell (0/25 cases), and its aggregate upper bound is only 0.1540. The original calibration table (display values) was 0.114655 for $\lambda = 0.76$, 0.115096 for $\lambda = 0.83$ and 0.115286 for $\lambda = 0.86$ — all of them unreachable, since the $n = 9$ ceiling is 0.669.

The route was formally downgraded after a calibration on $n = 25$, where the same method closes with effective universal bound 0.2235441655 (that is -6.52% from the exact value, or -5.75% on the most favourable reading); on $n = 100$ the same route is -9.34%, and its perimeter-defect constant is $\lambda \approx 0.15$ against a threshold of ≈ 1.01. The three lemmas used there were the boundary Euler accounting $\sum(6 - m_i) = 6 + 2b$, "boundary cell = one extra side" $\max_\Theta g = f(m+1)$, and the arc-covering bound $b \geq \pi/\arcsin r$; the first of these is precisely what Lemma 5 later replaced by the safer inequality $\sum_i e_i \leq 6N - 6 - B_{\mathrm{arc}}$. In other words, the modern universal bound of §3.3 is the repaired descendant of a route that was abandoned in its original form.

3.6.2 Witness calibration: correct but too weak

A finite-witness route was fully certified and then shelved. With 21 rational witness points, 32 maximal patterns, 53 feasible non-empty subsets and 506 enumeration events, total weight 21/2, it excludes $n = 8, 9, 10$ at $r = 7/25$. The independent checker passes 74 checks with 0 failures. But the bound is weaker than the known area bound for $n = 8, 9, 10$ and does not improve the $n = 100$ interval: on this grid the dual optimum at $r = 7/25$ is exactly 12, and for $r \geq 3/10$

eight disks already cover the witness set. The construction therefore cannot become a route past the area bound, and the analytic completeness of the finite reduction depends on a planar Helly / minimum-enclosing-circle support bound that is not formalized.

3.6.3 Simple global energies fail on the candidate graph

An angular/harmonic energy route was proposed to transfer the stress structure of an absorbed $n = 19$ proof to $n = 100$. Two certificates settle the matter. First, the true harmonic-extension stress energy has a quadratic form whose interval upper bound is strictly below $-1/100000$ at the candidate root in the candidate angular domain (`verify/out/n100-angular-energy-independent.json`, exit 0). Second, a rational rotation counterexample shows that inside the allowed angular domain the energy can *decrease* by more than 10^{-12} (`verify/out/n100-harmonic-energy-relaxation-counterexample.json`, exit 0). Hence the fixed-stress global convex-energy route is unavailable and must be replaced by per-rod force and constraint branching with quantitative isolation (`verify/out/n100-angular-energy-obstruction-audit.json`). This is a statement about one candidate contact graph; it is not a covering counterexample and it does not contradict the local second-order condition.

3.6.4 Two more numerical routes that were not promoted

- A per-cell equal-arc / edge-count-$5, 6$ sampling family found 0.1118772, but completeness of the adversary family was never proved, so it cannot serve as a universal bound.
- A radial-shell LP reported a value near 92.56; a fine-grid recheck found the sampled solution infeasible, so it is not a certified bound on the covering number. The report was corrected under independent challenge. This does not prove that finite-witness routes are impossible — only that the tested relaxations did not deliver.

Part IV. The six-branch engine: a contradiction at a fixed radius

4.1 The logical skeleton

Everything in this Part has the following shape.

1. Fix a rational *candidate radius* R and suppose, for contradiction, that 100 closed disks of radius R cover D.
2. **Normalize** every centre at that radius (same-radius normalization) and rebuild the ownership relation, the number of effectively used disks, and the common triangulation. Nothing from a previous radius may be reused.
3. **Count.** Derive necessary conditions that exclude all configurations with at most $m-1$ effectively used disks, and, for $m = 100$, restrict the number B of boundary arcs to a finite list.
4. **Split into branches.** The surviving (m, B) pairs are partitioned into six mutually exclusive branches $B31$, $B32$, $B33$, $B34$, $B35$, $B{\geq}36$.
5. **Close each branch** by an area/curvature budget: an exact rational inequality that contradicts coverage.

6. **Combine.** A combiner checks that every branch is closed at the same radius, that budgets are spent once, and that any branch closed at a *larger* radius is inherited through radius monotonicity.

The last point is a theorem, not a convention:

Lemma 6 (Radius monotonicity)**.** *If no covering by* 100 *disks of radius* R' *exists, then no covering by* 100 *disks of radius* $R < R'$ *exists.*

Proof. A covering at radius R is a covering at radius R' after enlarging every disk, since $B(c, R) \subseteq B(c, R')$. □

Hence a branch certificate established at $R' = 2831/25000$ is available at any $R \le R'$, provided the combiner records the inheritance explicitly (grade D evidence). The tail branch is *not* inherited in this way in the current chain: it is replayed directly at $R_\star = 22647/200000$.

4.2 Normalization at a fixed radius

4.2.1 The radial map

For a candidate radius R, define the radial map

$$d' = \frac{1 - R^2}{d},$$

where d is the distance from the origin to a centre, and let u be the unit vector of that centre. The identity

$$\|z - d'u\|^2 - R^2 = \frac{d'}{d}\big(\|z - du\|^2 - R^2\big) + \Big(1 - \frac{d'}{d}\Big)\big(\|z\|^2 - 1\big) \tag{1}$$

is asserted exactly by CAS in the certificates. Since $d'/d \le 1$ and $\|z\|^2 - 1 \le 0$ for $z \in D$, every point already covered at radius R remains covered after the map. The normalization is applied to *all* centres *before* the counting branch is taken; ownership, the number of effective disks and the common triangulation are rebuilt from scratch. This is the reason the engine cannot be reused at a new radius by editing one constant: the diagram itself changes.

4.2.2 The common counting bridge

At $R = 11323/100000$ the counting bridge excludes every configuration with at most 99 effectively used disks: $n = 99$, 96 cases, tightest $B = 34$, with the exact minimum area margin

$$\frac{935917323477479200061531074394942161076616337778844768733677817376904639496068264591580122993044 39}{17 \cdot 10^{100}} > 0$$

(stored in the external replay certificate). For $m = 100$ the surviving boundary counts are exactly

$$B \in \{31, 32, 33, 34, 35, 36, 37, 38, 39\}$$

(`complete_surviving_boundary_counts`). The six branches partition this list, and they are mutually exclusive by definition of B.

The same bridge was re-derived at $R = 2831/25000$: all $m \le 99$ are excluded, and for $m = 100$ only $B31$–39 survive. The normalized centre-norm-squared upper bound, i.e. an outward rational enclosure of $1 - R^2$, is

$$1 - R^2 \;\le\; \frac{9871789671}{10^{10}} \quad (R = 11323/100000), \qquad 1 - R^2 \;=\; \frac{616985439}{625000000} \quad (R = 2831/25000),$$

and at the accepted radius $R_\star = 22647/200000$ one has $1 - R_\star^2 = 0.987177834775\ldots$; the accepted certificates use the valid outward bound $98717894445399/10^{14}$ for the smaller radius $1132301/10^7$. These are the same quantity at different radii, and each is quoted with its own radius.

4.2.3 Branch vocabulary

Symbol	Meaning
B	number of boundary arcs served (equivalently boundary vertices of the common diagram)
m	number of effectively used disks
k	number of distinct boundary-serving disks
M	number of interior faces of the normalized common triangulation
e, b	number of ears, resp. bad faces ("$e0$", "$b0$" mean zero)
q	number of marked faces of a vertex; $q1$ is a "bad $q1$"
h	total degree excess of strictly deep vertices
C	number of chords of the core boundary
a	$C - f_3$, where f_3 is the number of empty core-boundary triangles
DDD, DDR, DRR, BRR, RRR	face types by the degrees (deep/ring) of their three vertices
core P/F	number of boundary vertices / faces of the induced core disk

The recurring mechanism in every branch is a *budget*: the boundary arc length available at radius R must pay for interior area, for boundary "marked" triangles, for ear and bad-face deficits, and for one hinge or variance payment. Two structural rules are enforced throughout: (i) a physical edge shared by two faces is paid *once*, and opposite half-edge terms cancel exactly; (ii) a scalar deficit or a local gain may not be spent in two different rows. The certificates record these as explicit fields (`physical_edge_pairs`, `hinge_payment = 0`, `variance_payment = 0`, budget-review strings).

4.3 Branch B31

$B31$ is the smallest boundary count. Its certificate is the "global full-face curvature" row: 40 face supports and 32 vertex supports, a degree window, in-place charges, ear and bad-face coefficients, and two M-intervals ($M \le 23$ and $M \ge 24$). Structural fields worth noting: all twenty face-label multisets are allowed, no deep-label assumption is made (`deep_label_assumption = false`), no reflected-centre hypothesis is used, the squared edge penalty and the half-side multiplier are both 0, and arbitrary ears and bad faces are allowed. The branch also carries an $n = 99$ exclusion with 96 cases and tightest $B = 34$, whose margin equals the counting bridge's.

The geometric skeleton transferred to a new radius is written out as obligations in the certificate itself: same-radius whole-cover normalization and diagram rebuild; the unconditional full-star degree-loss inequality and its budget comparison; whole-disk /6 and induced-core /5 isoperimetry; identical full-face labels with signed edge cancellation and a single budget use; and a closed-endpoint finite-type continuity argument.

Radius	minimum area margin	source
11323/100000	$379722800157300461201 1/10^{24}$	external replay certificate
11324/100000	$3205113070935301276991/10^{24}$	`n100-B31-at-11324.json`
$1132301/10^7$	$3791306959813284956819/10^{24}$	stage 143 margin table

4.4 Branch B32

$B32$ splits into $M \le 25$ and $M \ge 26$.

4.4.1 Low M: $81 = 73 + 8$ integer profiles

The enhanced low-M reduction excludes $M \le 24$, leaving $M25$ reduced to a unique deep degree-5 point, at most two high-degree points, and four cyclic star patterns. The original $M25$ fee row was then evaluated on all 81 integer profiles: 73 rows strictly positive, 8 negative. The 8 residuals are all of type $e0$ with two degree-7 high points, $p \in \{0, 2\}$ and central star type 2 or 3; the worst original gap is about 0.00012804, which is why looking only at the first failure (about 0.00002125) would have been misleading.

For this fixed integer domain the half-side multiplier, area ratio and perimeter penalty were retuned to

$$K = \frac{78619}{283293}, \qquad \epsilon = \frac{499143}{911596}, \qquad \tau = \frac{930249}{710647},$$

with the FAN payment strictly 0 because these rows have no $p3$, and with the normalized boundary envelope and the zero-slope hinge rechecked exactly at the new K (no payment is carried over from the old K). The exact receipts are

$$\min_{73 \text{ original rows}} = \frac{15600046212127187429}{2000000000000000000000000} > 0.00007800,$$

$$\min_{8 \text{ new rows}} = \frac{1596099436623121073}{1250000000000000000000000} > 0.00012768.$$

An independent `Fraction` ledger recomposes all 81 rows and recomputes 9 ordinary face classes, 18 outside face classes and the residual central faces; all strict lower endpoints are positive. The discovery-stage sentinel and K-objective errors were corrected, and a `baseline_numeric_area_margin` field that mixed up K was formally withdrawn by an erratum (`n100-discovery-erratum-145.json`); acceptance rests only on the exact certificate.

4.4.2 High M: the marked degree-5 point

The high-M diagnostic showed that the original face/vertex supports, the double-positive-curvature exclusion and the M/ear/bad-face shift checks all pass, while the marked-low-point exclusion alone still has margin

$$-\frac{490979625631490392613}{10^{24}}.$$

The response was to change the exhaustive classification rather than to accept the failing assertion. With all original shift inequalities retained, the Euler identity and the double-curvature exclusion leave only two possibilities:

1. $M26/e0/b0$ with all 68 interior points of degree 6;

2. $M26/e0/b0$ with a unique degree-5 point, a unique degree-7 point, and 66 points of degree 6; the marked count p of the degree-5 point is allowed to be $0, 1, 2, 3$.

The unique degree-5 point is then classified by whether it owns an original boundary neighbour:

- *No original-boundary neighbour*: then $p = 0$, the original direct star support applies, and the 4 height-marking counts times 2 high-neighbour star types give 8 rows with minimum margin
$$\frac{78693035940218423479}{500000000000000000000000} > 0.00015738.$$
- *Has an original-boundary neighbour*: then the degree-5 point cannot lie in the strict interior of any induced core face block, so planar separation and the /6 isoperimetric argument still apply. The core remains 68 vertices, 108 faces, a 26-edge chordless boundary and 42 deep points. Deep degree excess h is 0 or 1, and the four-face metric covers both cases with minimum margin
$$\frac{225914348193616561751}{500000000000000000000000} > 0.00045182.$$

The key angular ledger does not depend on p: with Γ the total marked interior angle sum, the ring-point occurrence count is $(72-h)+52 = 124-h$, the ring-point centred angle sum is $(32+h)\pi/3-\Gamma$, and the original-boundary centred angle sum is $\Gamma - 2\pi - 26\pi/3$; CAS asserts both identities and the merged linear-potential identity. The face counts
$$\mathrm{DDD} = 62 + h, \qquad \mathrm{DDR} = 20 - h, \qquad \mathrm{DRR} = \mathrm{BRR} = 26$$
also depend only on h. The 26 physical DRR/BRR half-edge pairs cancel, so no additional budget is spent.

4.5 Branch B33

4.5.1 The real residual: a unique degree-5 point

For $M \geq 26$, if a degree-5 point exists it is unique, of type $M26/e0/b0$, and the total high curvature is 2; hence the high points are either two degree-7 points or one degree-8 point. The classification is again by whether the degree-5 point has an original boundary neighbour — and the report is careful that $p = 0$ does *not* imply the absence of a boundary neighbour, so the two cases must be split by the neighbour, not by p.

4.5.2 No original-boundary neighbour: one coherent star replacement

Keeping the valid $K = 11/25$ and hinge slope $3/100$ and all face fees, the five central faces of the unique deep degree-5 point are replaced *once* by a coherent polygon support with
$$\epsilon = \frac{26304223}{69960624}, \qquad \tau = \frac{110526397}{84434654},$$
and a re-optimized central potential. With at most two high points the neighbour binary 5-cycle has only 4 counting types, checked by machine; the original 20 face supports, 18 outside supports and 8 central supports are recomputed exactly; no face contains three distinct high points. Every cyclic normalization gain is strictly greater than $9/100$, so the old negative degree-5 row plus $R^2 \cdot 9/100$ gives the conservative positive margin
$$\frac{185371504379126342167}{250000000000000000000000} \approx 0.000741486.$$
The five selected physical edges cancel opposite half-edge payments; no second hinge is spent.

4.5.3 Has an original-boundary neighbour: the labelled ring metric

Here the core is a simple chordless disk with $I67/P26/F106$ and 41 deep points. Since $P26 > 3$, every ring point has at least one deep neighbour and at least two core faces; a ring point also has exactly two outer-band BRR faces and p marked faces, so its original degree d satisfies $d-p-2 \geq 2$, i.e. $p \leq d-4$. This is a consequence of *this* core topology and imposes nothing on arbitrary interior points. The unique ring degree-5 point therefore has $p \in \{0,1\}$; ring degree 7 needs only $p \leq 3$ and ring degree 8 only $p \leq 4$.

With deep excess $h \in \{0,1,2\}$ the region face counts follow from

$$\mathrm{DDD} + \mathrm{DDR} + 26 = 106, \qquad 3\,\mathrm{DDD} + 2\,\mathrm{DDR} + 26 = 246 + h,$$

giving $\mathrm{DDD} = 60 + h$ and $\mathrm{DDR} = 20 - h$. The 26 core ring edges pair with the 26 outer BRR faces physical-edge by physical-edge. Enumerating the positions of 0, 1 or 2 high points around the 26-cycle and aggregating the five edge types RR/RH/HH/RL/HL together with ring degrees 7/8 and marked counts produces exactly 108 counting models. The selected rational parameters are

$$LR = -\frac{2217213}{59317159}, \quad LH = -\frac{47596282}{80611277}, \quad LL = \frac{20620389}{94559174}, \quad LB = -\frac{5596292}{14123693},$$

$$\tau = \frac{11510567}{31711524}, \qquad K = \frac{17082939}{31059893},$$

and 14 full-simplex supports (DDD, three DDR classes, five DRR, five BRR) are recomputed with every angular degeneracy bound and localization guard. All 108 rows are positive with minimum margin

$$\frac{147611772341741487969}{200000000000000000000000} \approx 0.0007380589.$$

The branch spends no hinge and no variance.

4.5.4 The no-degree-5 components

Three fresh subprocesses replay the $M26$ seven-face metric, the $M27$ with-chord metric and the one-ear metric. The old "$b1/e0$ impossible" and "$M26$ degree-7 cannot be deep" claims are re-examined through core connectivity/empty-side ear generation and the general /6 inner frontier inequality

$$26 + 19 + \frac{19^2}{6} < 106.$$

The final independent composition entry re-enumerates $M/e/b$/degree-5 counts, solves the DDD/DDR identities with CAS, matches all 26-cycle signatures to the 108 models, recomposes the 108 final area upper bounds from 14 signed supports with an independent `Fraction` ledger, and then accepts. It refuses to run under `python -O` and its negative control exits 1 with *Assertions must remain enabled.* The overall minimum reported positive margin of the whole branch is still set by the low-M part, about 0.00000294.

An earlier attempt at a uniform ring potential failed honestly: the optimal LR tended to 0 and the first two rationalizations left no strict lower bound; the recorded minimum predicted margin was about -0.00067010, and it was *not* accepted. Only the 14-face structure distinguishing low, high and ordinary ring points produced the exact positive certificate.

4.6 Branch B34

4.6.1 Low M: an isolated five-star

Two original endpoints were negative ($M24..25$ and $M \geq 26$); the diagnostic stage released only the $M24$ assertion to expose the second failure, keeping all 48 face classes, 1455 vertex classes, shift inequalities, normalized boundary, $q4$ payment and core premises actually passing. The low-M closure then uses the original vertex reduced costs and the negative M/ear/bad-face coefficients: $M22..23$ is positive; $M25$, or $M24$ with at least one ear or bad face, is positive; selecting one guaranteed deep low-degree point, any additional global degree-3/4/5 point's neutral vertex margin covers the negative area, and if the selected deep point is itself of degree 3 or 4 its own vertex margin also covers it. What remains is

$$M24/e0/b0, \quad \text{exactly one global degree-5 point}$$

$$\text{selected as the deep witness}, \quad \text{all other interior degrees} \geq 6.$$

Two peels then show the induced core of $I66/M24$ must be a simple $P24/F106$ disk (/5 excludes multiple components, bridges and multiple blocks), that the degree-5 point's neighbours are all strictly interior core points, and that if it were adjacent to a high point the high point would have to be the unique deep degree-7 point and the core would have to be a chordless simple $P18/F64$ disk — which the inequalities $18 + 12 + 12^2/5 < 64$ and, after one more peel, $18 + 13 + 13^2/6 < 64$ both contradict. Hence all five neighbours of the degree-5 point have original degree 6, and the five central faces have potential type $(P, 0, 0)$. Replacing those five faces *once* by the complete five-star cap

$$\mathrm{cap}_5 = \sqrt{\frac{25\,(25 + 11\sqrt{5})}{32}}$$

(with positive central angle sum 2π, giving angular potential term $P\pi/3$) yields the exact normalized gain

$$\frac{70784489700959821434439}{10^{24}} > 0.07078448,$$

and a final area margin

$$\frac{448474306824206627004168866279}{625000000000000000000000000000000} > 0.00071755.$$

An independent recomputation with $\sqrt{3}$ bounded below by $1732050807/10^9$, $\sqrt{5}$ bounded above by $2236067978/10^9$ and $\mathrm{cap}_5 < 6.23$ reproduces a positive margin

$$\frac{190482881108285619519923}{292968750000000000000000000} > 0.00065018.$$

4.6.2 High M: three topology slots

The $M26$ row's positive neutral low-degree vertex margins exclude any degree 3/4/5; the negative M slope and the ear/bad-face shift fees exclude $M \geq 27$, at least one ear, and at least one bad $q1$. What remains is

$$M26/e0/b0, \quad \text{all original interior degrees} \geq 6, \quad \text{positive degree excess } 2,$$

with the high points being two degree-7 or one degree-8 — and no assumption that they lie on the core ring. The core is $I66/P26/F104$ with 40 deep points; /6 excludes multiple components,

bridges and multiple loops, leaving a simple disk, but $P26/F104$ permits one chord, so the chordless conclusion of $B33$ cannot be copied. With C chords, f_3 empty core-boundary triangles, $a = C - f_3 \geq 0$ and deep excess h, the chord-aware first peel gives

$$b = 20 - h - C - 3a, \qquad \text{deleted faces} = 46 - h - C - a,$$

and /6 gives

$$104 \;\leq\; 46 - h - C - a + \frac{(20 - h - C - 3a)^2}{6}.$$

Machine exhaustion over $0 \leq h \leq 2$, $0 \leq C \leq 23$, $0 \leq a \leq C$ with non-negative frontier leaves exactly three slots:

core chords C	deep excess h	a	inner frontier	inner faces
0	0	0	20	58
0	1	0	19	59
1	0	0	19	59

The one-chord slot has exactly one empty core-boundary triangle, all deep points of original degree 6, and total high curvature 2 on the ring. The chordless $h = 1$ slot has one deep degree-7 and one ring degree-7; the chordless $h = 0$ slot has all high curvature on the ring. CAS caught that the general DDD formula had omitted $+a$; the failing original script is kept and the corrected counts are

$$\text{DDD} = 58 + C + h + a, \qquad \text{DDR} = 20 - C - h - 3a.$$

All three surviving slots have $a = 0$, so the correction does not change them. The one-chord support must retain DDD = 59, DDR = 19, DRR = 25, paired BRR = 24, unpaired BRR = 2 and RRR = 1; it is *not* legitimate to treat it as 26 fully paired DRR/BRR pairs.

The branch then closes with 23 chordless models and 47 one-chord models, both refined to a global boundary envelope of 10 000 interval boxes. The chordless models use 9 signed face supports with 26 physical edge pairs and zero hinge/variance payment; the one-chord models additionally distinguish 24 paired outer DRR/BRR edges, one chord edge, one RRR triangle and two unpaid BRR faces. Independent `Fraction` recomposition and CAS centred-angle identities pass. A uniform ring-potential discovery run for this branch recorded an optimal predicted margin of about -0.00103852; it was a discovery only and accepted no exclusion.

4.7 Branch B35

The diagnostic kept all faces, vertices, $q4$ payments, normalized boundary and topology premises, and showed negative area endpoints for the original $M \leq 24$ flagged rows and for restricted $M25$ and $M26$, while the $M \geq 27$, ear, bad-$q1$ and additional-low-degree shift margins remained positive. The strict residual was therefore

$$M \leq 24 \text{ flagged low-}M; \qquad M25 \text{ with all interior degrees} \geq 6 \; (\approx -0.0004727668);$$

$$M26 \text{ with all interior degrees} \geq 6 \; (\approx -0.0002967289).$$

It returned `status = PASS_PREMISES` with `B35_closed = false` — the state word that prevents a premise check from being read as a closure.

The closure then proceeded exactly as in $B34$: the low-M part is closed by the /5 contradiction between the unique unmarked degree-5 point and the number of core ears (at most one core ear

is allowed); $M25$ is closed by 8 ring models (core $P25/F103$, 40 deep vertices, 25 physical edge pairs, four ring-7 points with $pH \in \{11, 12\}$); $M26$ is closed by 3 chordless models plus 5 one-chord models (core $P26/F102$, 39 deep vertices, DDD = 56, DDR = $[20, 0]$). Each new metric has its own complete boundary-K and face sum with exact physical edge cancellation, and no scalar deficit is paid across two rows. The combined independent certificate reports `M25_closed = true`, `M26_closed = true`, `M_ge27_closed = true`, with model counts $\{M25 : 8,\ M26_{\text{chordless}} : 3,\ M26_{\text{onechord}} : 5\}$.

4.8 The tail branch $B \geq 36$

4.8.1 The capped-pair relaxation

For the high boundary counts the engine changes character. Instead of a global face ledger it constructs a *local pair loss* for each adjacent pair of boundary cells. In the absorbed formulation a pair term has the shape

$$\Phi(a, b) = \lambda \, \|c_a - c_b\|^2 + \frac{H(a) + H(b)}{2} - \frac{\mu \, (\arcsin s_a + \arcsin s_b)}{2},$$

where a, b are adjacent half-arc angles, c_a, c_b the corresponding centres, H a geometric loss built from the arc segment and the distance from the centre to the arc chord, and λ, μ penalty and slope coefficients. When the local terms are summed around the ring, the exact identity

$$\sum_i \frac{a_i + a_{i+1}}{2} = \sum_i a_i = \pi$$

is used — not a Jensen-type "equal arcs" conjecture. Each local pair upper bound is then fed into a total area budget, and two scalar checks are performed: the margin at $B = 36$ must be strictly positive, and the gain per additional boundary cell must be strictly positive, which extends the exclusion to every $B \geq 36$.

The scalar budget actually implemented by the certificate scripts is

$$\underbrace{(1 - \mu)\, \pi_{\text{lo}}}_{\text{available arc budget}} \; - \; \underbrace{\big((2m - 2 - B)\, A + B\, K\big) R^2}_{\text{interior and marked triangle area}} \; - \; \underbrace{B\, L}_{\text{intercept}} \; > \; 0, \qquad \underbrace{(A - K) R^2 - L}_{\text{gain per boundary cell}} \; > \; 0, \quad (2)$$

with $m = 100$, $A = 129903810568/10^{11}$ (an outward rational enclosure of $3\sqrt{3}/4$) and K the marked-triangle constant. Substituting the frozen constants reproduces the stored margins exactly, which is how the two certificate fields below were checked:

$$\text{at } R = \tfrac{113237}{10^6},\ L = \tfrac{75097}{10^7}: \quad \text{margin} = 4.906445222464608 \times 10^{-7},$$

$$\text{gain} = 5.1549192230158705 \times 10^{-3},$$

$$\text{at } R = \tfrac{22647}{200000},\ L = \tfrac{939}{125000}: \quad \text{margin} = 1.8087101250224213 \times 10^{-5},$$

$$\text{gain} = 5.152171860143971 \times 10^{-3}.$$

The underlying geometric derivation of (2) lives in the absorbed third-party package; what this project certifies is the exact arithmetic of the budget, the completeness of the rational square-domain subdivision described next, and the negative controls.

4.8.2 Rational square-domain subdivision

The pair domain is a two-dimensional rational square. It is covered by recursive rational box subdivision; for each box the local pair loss is bounded outward and compared with the budget. Boxes on the diagonal are skipped by symmetry and counted separately. Three "q-marked-side" triangle supports are verified independently. The accepted counts are

Certificate	radius	visited boxes	accepted boxes	symmetry-skipped
$B = 36$ at $R_\star = 22647/200000$, $L = 939/125000$	0.113235	1 841 089	917 297	—
$B \geq 37$ at $R = 2831/25000$, $L = 19/2500$	0.11324	93 257	46 038	591

with q-support box counts 3761, 104 and 35 for 1, 2 and 3 marked sides, and `pending_boxes` empty for $B \geq 37$. The exact margins are

$$B = 36: \quad \frac{1808710125022421182469778967931605940457120574226219}{10^{56}} \approx 1.8087 \times 10^{-5},$$

$$\text{per cell}: \quad \frac{2576085930071985489}{5 \cdot 10^{20}} \approx 5.1522 \times 10^{-3},$$

$$B = 37: \quad \frac{1664382916772173399824697789679316059404571205742262 19}{10^{56}} \approx 1.6644 \times 10^{-5},$$

$$\text{per cell}: \quad \frac{39572580329210081}{78125000000000000000} \approx 5.065 \times 10^{-3}.$$

The full run for $B = 36$ took 130.04 s of reported time; the tail is the most expensive part of the whole engine.

4.8.3 Why the tail does not extend to 0.11324

At $R = 2831/25000$ the *old* fixed coefficients give a total budget of about -0.000160907: the tail cannot close at 0.11324 even if its local supports remain valid. Holding μ and λ fixed, the intercept would have to be strictly below about 0.007505530351154887; the trial $L = 0.007505$ gave a conditional budget of about $+0.0000190926$, but the complete two-dimensional local support was never verified (a 250 000-box run accepted 124 491 leaves, left 17 pending stack items and returned `UNDECIDED`).

The obstruction is exact, not numerical. With the old constants, at the rational point

$$x = y = \frac{9921033}{12500000}$$

the pair relaxation satisfies

$$\frac{3754996055077}{500000000000000} \leq \text{pair} \leq \frac{3754996055079}{500000000000000},$$

so its lower endpoint exceeds the old intercept 1501/200000 by

$$\frac{2496055077}{500000000000000} \approx 4.99211 \times 10^{-6},$$

and it also exceeds the largest intercept that the $B = 36$ area budget can pay for,

$$\frac{270199092634633283031824697789679316059404571205742262 19}{360000000000000000000000000000000000000000000000000000000000000000000000},$$

by

$$\frac{160623330910716968175302210320683940595428794257737 81}{360000000000000000000000000000000000000000000000000000000000000000}.$$

Consequently the deficit cannot be repaired by increasing a box budget or by raising the intercept: the latter makes the $B = 36$ budget negative. A later exact diagnostic showed that both support terms vanish at the obstruction point, so refining the support boxes alone is mathematically insufficient as well. The obstruction certificate carries `status = PASS_OBSTRUCTION_TO_OLD_RELAXATION`, `full_B36plus_closed = false`, `global_bound_improved = false`, and its scope string states explicitly that this is not a covering configuration and not a counterexample to the target lower bound.

4.9 The combiner

The final assembly is deliberately thin, because everything of substance is inside a branch certificate.

- `verify/out/n100-candidate-complete-225.json` (schema `n100-candidate-complete/v1`, `status = PASS`) records `candidate_radius = 22647/200000`, `previous_target_radius = 2831/25000`, `closed_branches = [B31,...,B35,B36plus]`, `all_six_closed = true`, `global_bound_improved = true`, the monotonicity justification in prose, and for each of $B31$–$B35$ the source radius 2831/25000, the inherited radius 22647/200000 and the SHA-256 of the source certificate. The tail entry carries its own radius and hash.
- `verify/out/n100-six-branch-ledger-226.json` (schema `n100-six-branch-ledger/v11`) records `closed_branches`, `pending_branches = []`, `all_six_closed = true`, and a `scope` string that says in words that only the candidate radius is covered and that $R = 0.11324$ remains unproved because its $B36$ direct obstruction persists.
- `verify/out/n100-small-advance-controls-143.json` rejects two destruction cases: one that disables assertions, and one that mismatches the old radius.
- `verify/out/n100-radius-dependency-audit-143.json` traces 20 processes and 28 `sys.modules`-registered sources, records the radius seen by every verification function, and confirms that no other global R exists; anonymously loaded backends are bound by the full 626-file hash list.

4.10 Branch certificates at a glance

Branch	complete coverage range	established at	certificate
$B31$	$M \le 23$ and $M \ge 24$, arbitrary ears/bad faces	11324/100000	`n100-B31-at-11324.json`
$B32$	$M \le 25$ ($81 = 73 + 8$ profiles) and $M \ge 26$	2831/25000	`n100-B32-complete-149.json`
$B33$	low M; $M25$ coherent replacement; all high-M Euler residues	2831/25000	`n100-B33-complete-167-v2.json`
$B34$	$M22..23$, $M24..25$, $M \ge 26$ (23 chordless + 47 one-chord)	2831/25000	`n100-B34-complete-180.json`
$B35$	low M, $M25$, $M26$, $M \ge 27$	2831/25000	`n100-B35-complete-203.json`
$B \ge 36$	$B = 36$ and $B = 37..99$	22647/200000 (tail), 2831/25000 ($B \ge 37$)	`n100-tail-candidate-224.json`, `n100-tail-B37plus-210-a.json`

All branches share the same analytic dependencies, stated in every certificate and never hidden: common normalized centre triangulation; planar separation of an original-boundary neighbour from a strict core-block interior; general /6 disk isoperimetry and the induced-core /5 isoperimetry; core/outer-band decomposition; the complete-star/perimeter lemma; full-simplex signed face supports; and the once-only physical edge cancellation. The certificates contain integer exhaustion, CAS identities and exact rational interval arithmetic — they do *not* contain a formal planar embedding kernel, and every one of them says so.

Part V. The current frontier: how the bound moved, and where it stops

5.1 The accepted interval and its history

The currently accepted, machine-replayable statement is

$$\boxed{0.113235 \;<\; r_D(100) \;\le\; 0.1154469548855829}$$

with the lower endpoint established by the six-branch chain of Part IV at $R_\star = 22647/200000$, and the upper endpoint by the replayed explicit configuration of Part II. The relative gap $(U_\star - R_\star)/U_\star$ is about 1.92% (display value).

5.1.1 The lower-bound track

Lower bound (exact)	decimal (display)	route / evidence
1/10	0.100000	area only; `verify/out/lowerbound_n100.json` (L01–L20)
554570693/5000000000	0.1109141386	repaired integer-edge route; `verify/out/strengthened-lowerbound-n100.json`
$1111720153/10^{10}$	0.1111720153	sharpened envelope; `verify/out/n100-algebraic-bounds-bundle.json`
2787/25000	0.11148	long-chord quadratic deficit; `verify/out/n100-longchord-bundle-frozen.json`
11323/100000	0.11323	external six-branch replay; `verify/out/external-disk100-bounds-67ebf1c.json`
$1132301/10^7$	0.1132301	same skeleton, radius advanced; `verify/out/n100-new-lower-bound-143.json`
22647/200000	**0.113235**	six branches, tail replayed at the candidate radius; `verify/out/n100-candidate-complete-225.json`

The last two entries are different in kind from the first four. The configuration-free bounds are single analytic inequalities; the 0.11323 and later values come from the contradiction engine, and their improvement is measured in units of 10^{-7} to 10^{-5} because each advance requires re-closing every branch at the new radius.

5.1.2 The upper-bound track

Upper bound	decimal (display)	route / evidence
153/1250	0.1224	stage-1 strict certificate of the historical configuration; `verify/out/coverage_n100.json`
$1223433/10^7$	0.1223433	stage-2 server search, six candidates; `verify/out/pilot_n100_stage2_summary.json`
$1/\sqrt{67}$	0.1221694443563052	triangular lattice 97+3; `verify/out/lattice97_upperbound_n100.json`
18761/156250	0.1200704	cooperative round 1; `n100-cooperative-anchor-tight/replay.json`
$119891/10^6$	0.119891	refine; `n100-cooperative-refine-tight/replay.json`
$119877/10^6$	0.119877	polish; `n100-cooperative-polish-tight/replay.json`
$1154469548855829/10^{16}$	**0.1154469548855829**	absorbed external configuration; `verify/out/external-disk100-margin-67ebf1c.json`

The margin certificate is worth reading carefully, because it shows the improvement is exact rather than cosmetic. The frozen 100-centre rational configuration is verified to cover the whole closed unit disk with

$$\text{covering squared radius} \;\le\; \frac{1332799939235381342323027277241}{10^{33}} = 0.013327999392353813\ldots = U_\star^2,$$

by checking 128 436 interior triples, 9 900 boundary pairs, 100 antipodal pairs and 1 exceptional case (138 437 objects in total, exact integer arithmetic). The certificate records a *reference* bound $115446954885583/10^{15} = 0.115446954885583$ and this project's local bound $1154469548855829/10^{16}$; the difference is exactly $1/10^{16}$. So the accepted upper endpoint is a strict improvement of 10^{-16} over the reference value, not a restatement of it.

5.2 From 0.11323 to 0.1132301: the transplant and the first advance

Stage 139 switched the programme from interface consolidation back to radius advancement. It replayed the absorbed six-branch skeleton inside an isolated working copy (exit 0, 626 files bound,

source and copy hashes unchanged after the run), and produced a six-endpoint rational radius scan (0.11323, 0.113231, 0.113235, 0.11324, 0.11325, 0.1133) recording, for each endpoint, the counting bridge, the $B31$ fixed-fee rows and the tail's conditional total budget. Two facts emerged:

- At $R = 0.11324$ the 99-disk exclusion still holds, the necessary boundary counts for 100 disks are still 31..39, and the $B31$ final row, ear/bad-face signs, degree budget and fan necessary restrictions all pass. At $R = 0.11325$ the $B31$ fan premise no longer closes, so the whole branch fails there even though its final area margin is still positive.
- With the old tail constants the total budget allows radii only up to a threshold in

$$\Big[\frac{11323679462887}{10^{14}},\ \frac{113236794628871}{10^{15}}\Big),$$

  and at $R = 0.11324$ the old tail budget is about -0.000160907. Holding μ and λ fixed, the intercept would have to drop below about 0.007505530351154887.

The $B31$ arithmetic was then replayed explicitly at $R = 11324/100000$ (`n100-B31-at-11324.json`, exit 0, 40 face supports and 32 vertex supports, minimum area margin $3205113070935301276991/10^{24} \approx 0.003205113$), with the fixed dimensionless fees/potentials/support coefficients left unchanged and only the radius (and the radius metadata in the proposal) adapted. The certificate lists the geometric transfer obligations separately from the arithmetic replay, which is the honest way to record that "the arithmetic still closes" is not the same as "the geometry still applies".

Stage 143 then advanced the radius to $1132301/10^7$ and re-closed all six branches with the original coefficients untouched. The exact same-radius margins were

Branch	minimum area margin at $R = 1132301/10^7$
$B31$	$379130695981328495681 9/10^{24}$
$B32$	$4005242817864335063 1/125000000000000000000000$
$B33$	$141216899554184842489/10^{24}$
$B34$	$57252253086154121097/250000000000000000000000$
$B35$	$37566017111533099749/500000000000000000000000$
$B \geq 36$	$336050926141338909263097789679316059404571205742262 19/10^{56}$

with the tail visiting 3 397 109 boxes and accepting 1 694 095, and with the per-additional-B margin strictly positive. The new normalized centre-norm-squared upper bound was $98717894445399/10^{14}$; the counting bridge at the same radius excluded all $m \leq 99$ and left only $B31$–39 for $m = 100$. The improvement is exactly $1/10^7$, and the stage is explicit that it used neither the stage-141 $M24$ reinforcement nor any change to the hinge slope, fee coefficients or support rows.

5.3 Closing the branches at 0.11324, one by one

Between stages 144 and 211 the six branches were re-closed at the harder target $R = 2831/25000 = 0.11324$, one at a time, with the failures kept on the record.

Stage(s)	Branch	Outcome
144–149	$B32$	complete: $M \le 25$ ($81 = 73 + 8$ profiles) and $M \ge 26$; two genuine failures found and repaired (a mixed-up K in a discovery field, withdrawn by erratum; and a marked-low exclusion that had to be replaced by a new classification rather than accepted)
150–168	$B33$	complete: low M, the coherent deep-5 star, and 108 labelled ring models; the first uniform ring potential failed with a predicted margin of about -0.00067010 and was not accepted
169–185	$B34$	complete: low M (isolated five-star) and high M (23 chordless + 47 one-chord models); two negative endpoints had to be exposed by diagnosis before they could be fixed
186–211	$B35$	complete: low M, $M25$ (8 ring models), $M26$ (3 chordless + 5 one-chord), $M \ge 27$
209–211	$B \ge 37$	complete at $R = 2831/25000$ with intercept $19/2500$: 93 257 visited, 46 038 accepted, 591 symmetry-skipped, 0 pending
206–229	$B = 36$	**not closed**: exact obstruction, then a conservative candidate that does close at a smaller radius

This is the point at which the accepted bound became 0.113235 rather than 0.11324: five of the six branches are available at the harder radius, but the tail is not, and a chain is only as strong as its weakest branch.

5.4 The $B = 36$ obstruction, and why the obvious repairs fail

5.4.1 The exact obstruction

At $R = 2831/25000$ with the frozen constants ($\lambda = 3/50$, $\mu = 92531/10^7$, marked constant $K = 973/3125$, old intercept $L = 1501/200000$) the rational point

$$x = y = \frac{9921033}{12500000}$$

has a pair relaxation whose interval is $[3754996055077/5{\cdot}10^{14},\ 3754996055079/5{\cdot}10^{14}]$. Its lower endpoint exceeds the old intercept by $2496055077/5{\cdot}10^{14} \approx 4.99211 \times 10^{-6}$ and exceeds the $B = 36$ area-allowed maximum intercept by $160623330910716968175302210320683940595428794257737 81/(36 \cdot 10^{55})$. The obstruction certificate therefore returns `PASS_OBSTRUCTION_TO_OLD_RELAXATION` with `full_B36plus_closed = false` and `global_bound_improved = false`, and its scope string states that this is not a covering configuration and not a counterexample to the target lower bound.

5.4.2 Four repairs that do not work

1. **More boxes.** The obstruction is a point-wise inequality at an exact rational point; the pair value there is what it is, so subdivision cannot lower it. A later exact diagnostic confirmed that both support terms vanish at the obstruction point (`n100-tail-point-branch-212.json`, `n100-tail-direct-components-216.json`), so refining the support boxes is mathematically insufficient.

2. **A higher intercept.** Raising L makes the $B = 36$ budget negative, because the budget already fails at the old intercept.

3. **Reusing the $B \geq 37$ intercept.** The $B \geq 37$ certificate closes with $L = 19/2500$; using it at $B = 36$ makes that budget negative.

4. **Synchronized retuning of slope and penalty.** A broader discovery search over the direct pair/angle family produced the optimum

$$\lambda = 0.0594469773645484, \quad \mu = 0.010360567875057177, \quad K \approx 0.31268174443157726,$$

with a local peak of about 0.00739518789840696 against a $B = 36$ allowed intercept of about 0.007391936425058569: a per-cell deficit of about $-3.25147334839 \times 10^{-6}$ and a total $B = 36$ deficit of about $-1.17053040542 \times 10^{-4}$. This agrees with an independent search in the same family. So within the frozen direct pair/angle relaxation, simple synchronized retuning cannot close $B = 36$.

5.4.3 What was actually tried and stopped

Two rational candidates were started and did not finish:

- $R = 113237/10^6$ with $L = 75097/10^7$: the scalar budget is positive (margin $4.906445222464608 \times 10^{-7}$, gain $5.1549192230158705 \times 10^{-3}$), and a 101-point rational scan between $1132301/10^7$ and $2831/25000$ showed that the known obstruction point stays within budget up to the exact sample $R = 11323703/100000000 = 0.11323703$ and strictly exceeds it from the exact sample $R = 113237129/1000000000$ onwards (30 of the 101 samples are obstructed by that one point). The scan's own scope string says it is diagnostic only: a sample not obstructed by this point is not a global $B36$ certificate. The full pair-domain replay near the budget edge subdivided for too long and was stopped.
- $R = 113236/10^6$ with $L = 0.007511$ and, more conservatively, $L = 0.00751$: both stopped for the same reason.

Neither is reported as a failure and neither is a certificate. The conservative candidate $R = 22647/200000 = 0.113235$ with $L = 7512/1000000 = 939/125000$ has a much wider budget margin and did complete: 1 841 089 boxes visited, 917 297 accepted, the three q-supports passing with $3761/104/35$ boxes, and both the $B = 36$ margin and the per-B gain strictly positive. That is the candidate which was accepted, and it is the reason the frontier sits at 0.113235.

Remark 8 (The undecided window, stated precisely). The verifiable undecided window is $(0.113235, 0.11324]$. Inside it, the fixed-old-parameter single-point scan indicates that from 113237129/1000000000 onwards that particular obstruction point alone blocks the old relaxation — but this is a statement about one relaxation at one point, *not* a mathematical upper bound on all new routes.

5.5 Three hard boundaries that the route catalogue respects

The systematic catalogue of Parts VI makes three boundaries explicit, and they are worth stating before the catalogue itself:

1. The recursive inner-disk counts (≥ 56, then ≥ 58 inner disks, hence $k \leq 44$ and then $k \leq 42$ boundary-serving disks) are consequences of a counting argument, *not* new global radius bounds. Throughout all of those stages the global interval was unchanged.

2. The progression $0.11516023 \to 0.11521651$ is a threshold advance *inside* the conditional branch $k \geq 55$. The complementary branch $k \leq 54$ was never excluded along that line, and one stage even produced an explicit integer feasible point of the aggregate relaxation (34 near-end + 20 tail-end + 46 non-serving = 100 disks with 54 serving), which forbids any logical consequence of that relaxation from being empty.

3. Every `PASS` in the interface/contract stages means contract registration, replay consistency or a timing sample — never a geometric closure. All 174 legacy sources are `PRECONDITION_MISSING`, and the dependency gate is `UNDECIDED`.

5.6 What a genuine advance would require

The frontier is not blocked by compute; it is blocked by the strength of one inequality. The certificates and diagnostics point to the same conclusion:

- **A new geometric inequality for $B = 36$**: a common-endpoint lens constraint, or a genuinely coupled multi-edge averaging inequality, replacing the direct pair/angle relaxation. A rewrite that merely re-expresses a term already present in the pair expression adds no information — this was tested explicitly (replacing a coarse $4\lambda R^2$ distance cap by the exact centre distance gives nothing new, because that term is already in the original expression).
- **A resumable pair-box queue**: persist the pending boxes per round, bind parameters by hash, cache inverse-trigonometric and local-support intervals, and shard the domain into diagonal and off-diagonal regions with multiple processes. The stopped runs at 0.113236 and 0.113237 failed on wall-clock, not on mathematics; without a queue, each new candidate restarts from the root.
- **A sharper universal bound**, if the engine is to be bypassed: the configuration-free best is 0.11148, which is 1.55% below the accepted engine bound, so this is not a near-term substitute.
- **The exact-value route**: a complete enumeration of the internal map family, a treatment of $m < 100$, degenerate configurations, and a closure of both bounds onto the same algebraic number, followed by irreducibility and root isolation. The square-relaxation / real-quantifier-elimination theory absorbed from the third-party material is the only known terminating route to an exact value, but it has no working $n = 100$ backend: the zero set of the compiled polynomial system is singular everywhere, so a regular Krawczyk box cannot replace a full QE backend, and the variable budget (605 geometric + 1318 relaxation = 1923, with 1922 eliminated) is a budget, not a runtime.

Part VI. The route catalogue: every family, its construction and its status

This Part is the systematic inventory the rest of the document prepares for. Each row gives the route, its construction, its status and its evidence. Status vocabulary: **closed** (a strict certificate exists inside its declared domain — possibly a conditional sub-branch), **partial** (genuine strict exclusions covering only part of the declared domain), **diagnostic** (measurement, audit or bottleneck only), **falsified** (the claim or its premise was explicitly refuted or a counterexample exists), **undecided** (ran to completion with zero new exclusions), **interface** (`PASS` means contract registration or replay

consistency, *not* geometric closure), **configuration** (an upper-bound construction). Unless stated otherwise, every route in this Part carries `machine_verified_geometry = false` and `global_optimality = false`, and the global interval quoted by the reports of that era is $11323/100000 < r_D(100) \le 1154469548855829/10^{16}$. Artifact paths are given relative to the repository root; see Part VII for which of them live in the published mirror.

6.1 Combinatorial and topological routes

The purpose of these routes is to turn a continuous covering into a finite combinatorial object: a planar map, a dart map, a triangulation with a fixed face catalogue, or a boundary word.

- **Planar / dart-map compiler.** vertices with independent edge identities, face permutations, ordered four-axis markings; half-edge code (α, ϕ) keeps multiple-edge identity; bounds $V \le 2m{+}2$, $E \le 3m{+}1$; model budget $k_M = 19m{+}22$ (1922 at $n = 100$) *Status: diagnostic.* Evidence: `verify/n100_generic_map_compiler.py`, `n100_dart_map_canonicalization.py`, `n100_raw_dart_model_adapter.py`
  Map canonicalization

6.2 Boundary-circle, axis and box-pruning routes

This family attacks the configuration space directly: propagate interval constraints on centre coordinates, split boxes, and count how many disks can serve the boundary.

- **Capacity frontier.** capacity / Hall-type necessary conditions on boundary service; box-aware pruning; the non-separated three-point capacity cut *Status: partial.* Evidence: `docs/result-n100-stage25-capacity-frontier.md`
  Box-aware common-centre pruning

6.3 Arc-pair, nogood and learning routes

This family works on one fixed core and one narrowed parent domain, learns necessary relations between boundary-tile labels, and asks how much support remains.

- **Arc-pair consistency and cycle support.** necessary pair relations between arc tiles sharing a centre (vertex-box minimum distance $\le 2U$); 289 530/277 368 pairs reduced to 1866/2312 labels; exact support of the inherited finite pairwise model via an independent-set DP on the 32-vertex cycle *Status: undecided.* Evidence: `verify/out/n100-arc-pair-consistency.json`, `n100-arc-cycle-support.json`
  Ordered and monotone cycles

6.4 Lens, duality, recursive-count and interface routes

- **Refined lens upper bound.** unified enclosure of the intersection lens at $d_0 = 1 - r$, with a per-strip upper Darboux sum of $\min(2rb{-}b^2,\ s^2{-}(1{-}2r{+}a)^2)$: 256 strips give 0.11521249, 4096 strips give $11521651/10^8$ ($\delta = 101/10000$), raising the certified $k \ge 55$ radius from 0.11516023 to 0.11521651 *Status: closed (conditional branch).* Evidence: `docs/result-n100-stage94-refined-lens.md`
  Joint support (superseded)

6.5 Energy, force and local-exclusion routes

- **Fixed-stress harmonic angular energy.** $E(q) = \sum c_{ij}\|q_i - q_j\|^2$ with $v^{\mathsf{T}} H_{\mathrm{ang}} v = 2\sum c_{ij}(q_i \cdot q_j)(v_i - v_j)^2$; a Loewner bound was hoped to give convexity on the angular domain *Status: falsified.* Evidence: `verify/out/n100-angular-energy-independent.json`, `n100-angular-energy-negative-witness.json`
  Angular energy, two exact obstructions

6.6 Radius-scanning and engineering routes

- **Six-endpoint radius scan.** exact rational scan at 0.11323, 0.113231, 0.113235, 0.11324, 0.11325, 0.1133 recording the counting bridge, the $B31$ fixed-fee rows and the tail's conditional total budget; at 0.11325 the $B31$ fan premise already fails *Status: diagnostic.* Evidence: `verify/out/n100-lower-radius-bottleneck-firstpass.json`
  101-point rational scan

Part VII. Reproduction, artifacts and dependencies

7.1 Entry points

7.1.1 The publication consistency check

The single cheapest entry point is

```
python reproduce.py          # from the repository root
```

It loads the four core certificates, requires their statuses, requires `candidate_radius = 22647/200000`, `accepted_global_lower_bound = 22647/200000`, `all_six_closed = true` and `pending_branches = []`, requires the string "0.11324" to remain present in the README's evidence-boundary section, and prints SHA-256 for each certificate. It does *not* re-run the historical recomputations, and it says so.

Its output in this audit was

```
n=100 publication consistency: PASS
accepted bound: r_D(100) > 0.113235
0.11324: not closed; B36 residual retained
```

followed by one line per certificate. The certificate names and their SHA-256 digests, as printed (wrapped here only for page width), were:

- `verify/out/n100-candidate-complete-225.json`
  `c58c71562ed014c0f716f02137dcf473d98fd73e6888f89aa9d06f08a41707a5`
- `verify/out/n100-tail-candidate-224.json`
  `b5ca9c60be73136356c5c77a3b13a9d9c22a07d265299cb3acda2262a5715fee`
- `verify/out/n100-candidate-budget-222.json`
  `41c0106a015c31c7a323639368cd9ff241c5d745c8a4f842356647dd4922d0e2`

- `verify/out/independent_lattice97_upperbound_n100.json`
  `c9f07ee0be9705cea9349fba2023af3a70824ee791373910809c802b2b1b4f4d`

7.1.2 Re-running individual routes

Each route has its own entry point, and every entry point refuses to overwrite an existing output target, so a genuine re-run needs a fresh path. Representative commands, as recorded by the reports:

Command	Route
`python verify/lattice97_upperbound_n100.py`	triangular-lattice upper bound
`python verify/independent_lattice97_upperbound_n100.py`	independent upper-bound replay
`python verify/run_algebraic_bounds_n100.py -out <fresh>`	sharpened envelope + lattice candidate
`python verify/run_strengthened_lowerbound_n100.py -out <fresh>`	integer-edge-count bound
`python verify/run_longchord_n100.py -out <fresh>`	long-chord bound
`python verify/run_n100_delivery_audit.py -run-dir <fresh>`	seal & fast replay
`python verify/run_n100_interface_suite.py -run-dir <fresh>`	nine-batch interface replay
`python verify/run_n100_boundary_line_replay.py -run-dir <fresh>`	$m99$ boundary line
`python verify/run_n100_full_revalidation.py -run-dir <fresh>`	all 4776 configurations
`python verify/run_n100_interface_timing.py -run-dir <fresh>`	timing calibration
`python verify/run_n100_composition_scaling.py -run-dir <fresh>`	composition & scaling
`python verify/run_n100_shared_support_timing.py -run-dir <fresh>`	shared-support session

Three of these (timing, composition/scaling and shared support) write to non-overwritable final targets and were *not* uniformly output-isolated; the integrated report records this as an outstanding wrapper debt rather than hiding it. Branch-level replays are driven by `verify/n100_B31_new_radius.py`, `n100_B32_complete_149.py`, `n100_B33_complete_167_v2.py`, `n100_B34_complete_180.py`, `n100_B35_complete_203.py`, `n100_tail_B37plus_resumable_210.py`, `n100_tail_candidate_224.py`, `n100_candidate_complete_225.py` and `n100_candidate_ledger_226.py`. The $B33$ and $B34$ entry points require assertions to be enabled: they reject `python -O` and report *Assertions must remain enabled.*

7.2 Artifact coverage: what is actually in the published mirror

This document was written against the working repository. Because a reproducibility claim is only as good as the bytes it points at, an explicit audit of artifact coverage was run for this monograph, and its result belongs in the paper.

Finding	Detail
Mirror is a subset	Of 100 artifact paths referenced by the mirror's own README, status file and stage reports, only 19 exist under the published mirror `n100-public/`; 81 are absent there, and 80 of those 81 do exist in the parent working repository (one probe path was an invented name and is excluded from the count)
`proofs/` missing	`n100-public/proofs/` does not exist, although both lattice certificates reference `proofs/n100-lattice97-algebraic-upperbound.md`. That file exists in the parent repository and its SHA-256 `73d39456b70fdcae41ede5718a608d139661a6c7f2a626cf95b4e348be19600b` matches the `proof_sha256` field recorded by the main lattice certificate
Transplant package	`runtime/disk100-release-lf-67ebf1c/` is absent from the mirror but present in the parent repository, including all four sources cited by the branch certificates (`verify_B31_global_low_degree_0p11323.py`, `verify_capped_pair_B36plus_0p11323.py`, `pair_bound_certificate_20261004/verify_pair_bound.py`, `threshold_11305_components/verify_remaining_cases.py`). A replay from the mirror alone therefore cannot execute the branch checkers
Intermediate certificates	The mirror keeps the final composition certificates and the tail certificates, but many intermediate sub-certificates (for example the $B32/B33/B34/B35$ component receipts and the stage-143 assembly receipts) are absent; their numbers can then only be quoted from the stage reports, not re-read from JSON
Root-only artifacts	`verify/out/external-disk100-local-quadtree/replay.json` and `verify/out/external-disk100-strict-local-67ebf1c.json` are referenced by the mirror's status file but exist only in the parent repository

Remark 9 (Consequence for this document). Every quantitative claim in Parts II–VI was checked against the artifact that exists in the parent repository, not against the mirror. Where a number could only be taken from a stage report because the underlying certificate is not in the mirror, the text says so or quotes the report. This is the same discipline the programme applies to its own stages, applied to the publication layer.

7.3 Binding inconsistencies found by this audit

Three binding issues were found and are recorded here rather than silently repaired.

1. **The independent lattice receipt binds an earlier generation.** The independent replay certificate `verify/out/independent_lattice97_upperbound_n100.json` records `input_sha256 = 7e732e189a0d69e30184dbd053ae2eda6ee588f0418d49553d71004e5f4ff475` and `proof_sha256 = a0454013448a2612475975b539d64fa722ab9e69434ca0d1d913bfba03d57d4b`, whereas the current main certificate file hashes to `7a98772a8fb4c608ca32ffe6489bf1f09006dd3c59a4322125beca8c1e72cd7d` and the current proof file to `73d39456b70fdcae41ede5718a608d139661a6c7f2a626cf95b4e348be19600b`. Its `checker_sha256 = 9655d038a2194ebf2f849ad550ccc645f10c5f7516270d7dfe47267cb23ac878` *does* match the current independent script. So the checker binding is current while the input and proof bindings are stale: the independent replay was made against an earlier certificate/proof pair. This does not invalidate the independent replay of the *claim* (the 17 checks pass on the current inputs when re-run), but the receipt as stored is not a same-generation binding, and it must not be presented as one.

2. **A referenced certificate path is missing from the mirror** (see the artifact-coverage subsection of Part VII).

3. **One documented constant is not in the published tree.** The tail budget uses a quantity imported from the absorbed package; the local re-expression of the budget in `verify/n100_candidate_budget_222.py` uses $A = 129903810568/10^{11}$, and substituting it together with the frozen slope, penalty and marked constant reproduces both stored margins exactly (see the budget re-check in the Appendix). The identity of that constant with an outward rational enclosure of $3\sqrt{3}/4$ is an *inference* from the arithmetic, not a quotation from the package.

7.4 External material and the absorption boundary

Source	Role and boundary
`cjt42554-crypto/disk100-covering-certificates` @ `67ebf1c5776f72a7b7786a8655eed5463429ae55`	supplies the replayed 100-centre configuration and the six-branch skeleton; 625 files, 7 269 413 bytes; the integrity block's own scope is "Release integrity only; mathematical results require replay". Its `not_claimed` list is explicit: global optimum, minimal polynomial of $r_D(100)$, formal proof-assistant geometry
`VonEquinox/DiskCoveringSolve` @ `0895849`	read, replayed and independently cross-checked: the $n = 19$ double chain and the $n = 20$ chain were re-run in isolated copies with exit 0; $n = 11..18$ were *not* replayed; no third-party source is redistributed by this project
`VonEquinox/DiskCover`	the HTTP request returned 404 and Git reported "Repository not found"; nothing from it is attributed or accepted

The absorption report states the transfer conditions for each reusable ingredient — complete topological peeling and safe pruning; exact force, Farkas and energy certificates; candidate-anchor isolation with a continuous-domain proof tree; a linear-size semi-algebraic cover model; the single-quartic square relaxation with real quantifier elimination; and exact endpoint polynomials with isolation. In each case the condition is that $n = 100$ must re-prove the family range, the necessary conditions and the recursive completeness: $n = 19/20$ certificates cannot be reused as $n = 100$ certificates.

7.5 Negative-control inventory

A certificate is accepted here only if its destruction tests are rejected. The battery in use includes, across the routes: a scaled-up target disk (precondition only, explicitly not an uncovered witness for the delivered claim); an explicit uncovered point for an enlarged-disk variant; a shrunk and an enlarged covering radius; a wrong metric (the missing factor 3 in the hexagonal inner product); a rotated and an inflated hexagon; a truncated neighbour search; a misattributed neighbour; two removed tangency centres; a removed interior centre; the 97-centre configuration submitted as a 100-point claim; a single shell centre removed (coverage still holds — only the cardinality is rejected); a broken dart pairing; a broken face permutation; a support declared inactive; an assertion-disabled run; a mismatched old radius; ten corrupted rational payloads; and 8 tamper tests on the local Krawczyk witness (zero inverse row, zero Cholesky transform, duplicate pivot column, non-dyadic inverse denominator, wrong witness dimension, negated Lagrangian Hessian, witness-source SHA mismatch, altered geometric polynomial source). Two entry points additionally refuse to run under `python -O`.

7.6 Environment

The published runs use Windows 11 with Anaconda Python 3.13.5 (`sympy` 1.13.3, `mpmath` 1.3.0, `gmpy2` 2.2.1, `python-flint` 0.9.0), and `tectonic` for document building. The decision layer needs only the standard library: `fractions`, `decimal`, `math`, `json`, `hashlib`, `pathlib`, `itertools`. Symbolic work uses `sympy` for identity checking and factorization, never as a substitute for a numerical certificate. Server-side discovery ran in a `crystalx` environment with no environment installation; the six-candidate server search completed 916 objective evaluations in 15.746 s and recovered 25 files whose SHA-256 all verified. The one external-project replay required only a Windows compatibility layer (path separators, executable names, LF output); the mathematical conditions, constants, intervals and counts were unchanged.

Part VIII. Open obligations, stated without softening

8.1 What is established, precisely

For the avoidance of doubt, the strongest statements this dossier can support are:

1. **An upper bound.** There is an explicit 100-centre rational configuration whose covering radius is at most $U_\star = 1154469548855829/10^{16} = 0.1154469548855829$, certified by 138 437 exact integer objects (128 436 interior triples, 9 900 boundary pairs, 100 antipodal pairs, 1 exceptional case), with the exact covering squared-radius bound $13327999392353813423230272772 41/10^{33} = U_\star^2$, improving the reference value $115446954885583/10^{15}$ by exactly $1/10^{16}$.

2. **A lower bound.** $B31$–$B35$ are excluded at $R = 2831/25000$ by exact certificates; $B \geq 37$ is excluded at $R = 2831/25000$ with intercept $19/2500$; $B = 36$ is excluded at $R_\star = 22647/200000$ with intercept $939/125000$ over 1 841 089 rational boxes. Radius monotonicity then gives $B31$–$B35$ and $B \geq 37$ at $R_\star$ as well, so

$$r_D(100) > 22647/200000 = 0.113235.$$

3. **Configuration-free bounds.** $r_D(100) \geq 2787/25000 = 0.11148$ (long chord), with $\geq 1111720153/10^{10}$ (sharpened envelope) and $\geq 554570693/5000000000$ (integer edge counting) also valid.

4. **An exact obstruction.** With the frozen capped-pair constants, the $B = 36$ relaxation at $R = 2831/25000$ exceeds both the old intercept and the area-allowed maximum intercept at the rational point $x = y = 9921033/12500000$. This is a statement about a relaxation, not about coverings.

   Everything else is either conditional, diagnostic, interface-level, or open.

8.2 The open obligations

Obligation	Precise statement of what is missing
Exact value of $r_D(100)$	not obtained. The accepted interval has width about 0.0022; no argument closes it. `global_optimality_proved = false`
Minimal polynomial of $r_D(100)$	not obtained: not the coefficients, not the degree, not the irreducibility, not a rational root-isolating interval. What exists is the minimal polynomial $67t^2 - 1$ of *one configuration's* covering radius, which is a different object and is explicitly scoped as such
Closure onto one algebraic number	the two endpoints are different numbers; nothing identifies the ideal envelope's zero with the optimal radius, and the envelope's zero is explicitly a statement about the method's limit, not about $r_D(100)$
Complete internal map enumeration	the only known necessary count domain is 87 (m, B) pairs with $m = 96..100$ ($96 : B32..35$; $97 : B29..42$; $98 : B28..46$; $99 : B28..50$; $100 : B28..54$), obtained under the accepted normalized-geometry reduction. It is not a complete topology enumeration, and the surviving families still contain infeasible graphs
$m < 100$ and other graphs	the counting bridge excludes $m \leq 99$ at the radii where it was replayed, but the general treatment of $m < 100$, of other graph families, and of degeneracies is not closed
Degeneracies	coincident centres and redundant disks are handled by normalization; the remaining degeneracy catalogue (tangential contacts, collinear centres, boundary multi-activity) is handled inside individual branches by explicit arguments, not by a single uniform machine-checked treatment
The $B = 36$ inequality	the direct pair/angle relaxation cannot close 0.11324; a common-endpoint lens constraint or a genuinely coupled multi-edge averaging inequality has been identified as the missing ingredient but has no passing global certificate
Resumable tail infrastructure	the 0.113236 and 0.113237 attempts stopped on wall-clock, not on mathematics; a resumable box queue with cached inverse-trigonometric and support intervals and diagonal/off-diagonal sharding is designed but not delivered
An exact-value backend	the absorbed square-relaxation / real-quantifier-elimination theory is the only known terminating route to an exact value, but it has no $n = 100$ backend; the compiled system's zero set is singular everywhere, so a regular Krawczyk box cannot replace a full QE backend, and the variable budget (605 geometric + 1318 relaxation, 1922 eliminated) is a budget rather than a runtime
Legacy source activation	all 174 legacy conditional sources return `PRECONDITION_MISSING`; the dependency gate is `UNDECIDED`. "Every historical variant is directly callable and individually re-timed" is *not* achieved
Output isolation debt	the timing, composition/scaling and shared-support entry points still lack a uniform repeatable-output wrapper; the integrated report lists this as owed work
Geometric formalization	no Lean/Coq kernel verification exists. Every universal statement is analytic prose plus exact finite arithmetic; the analytic/geometric bridge is a human-reviewed dependency, recorded as `machine_verified_geometry = false` in every certificate
Unpublished constants	at least one tail constant is imported from a package that is not part of the mirror; its value is re-expressed locally and reproduced numerically (Appendix C), but the identification of that constant with an outward enclosure of $3\sqrt{3}/4$ is an inference
Notation debt	the reports do not give a self-contained definition of the counting parameter M; it appears only through integer excess identities and surviving profiles. A reader must reconstruct it from the certificates, and this monograph flags that rather than inventing a definition

8.3 Claims that were withdrawn or corrected

A research record is only credible if it keeps its retractions. The following claims were made and then withdrawn or corrected, and the corrections are part of the record:

- **The $4\arcsin(r/2)$ arc bound.** Wrong; the correct maximum angular measure of $\partial D \cap B(c_i, r)$ is $2\arcsin r$, attained at $\|c_i\| = \sqrt{1-r^2}$. The wrong constant cannot imply $< \pi$ near $r = 1$ and is unusable.
- **The boundary-annulus "global no-gain" acceptance.** Voided: a wrong inner-face filter made the pending set empty, and the independent script had duplicated the same error. Exit code 0 is not evidence that the coverage domain is right.
- **The stage-47 v2 exclusion of all 4776 configurations.** Rejected.
- **The angular-energy inertia claim $+25/-6$.** Retracted: CAS on the real data gives one negative direction in the 32×32 angle coordinates (plus one zero = global rotation), and one negative after projection to the 31-dimensional gap coordinates. No inertia claim is made any more.
- **"The radial-shell LP gives a certified covering-number bound near 92.56."** Corrected: the sampled solution is infeasible on a 4001-point fine grid (coverage $0.99903 < 1$, dual $1.00317 > 1$); the only certified statement is the coarse bracket $[1/r^2,\ \pi/\mathrm{lens}(1, r)] = [80.911, 165.734]$ at $r = L$. The stronger phrasing "the route is closed / a certificate is impossible" was also withdrawn.
- **"The configuration side still has 5–6% of room."** Withdrawn: the small-n optima are ring-type families while the best $n = 100$ configuration is a lattice-type family, so they are not comparable; the lattice-family ratio $c(n)$ is 2.13 ($n = 40$), 1.92 (50), 1.35 (70), 1.14 (100), 1.12 (140), 0.82 (200).
- **Stage-142 discovery numbers** (an ear approximation of 1.332634 against a true 1.42016417919847, and a K-objective erratum) must not be quoted as certificate values.
- **The stage-44 first version** (4496 contractions / 597 incidences) is superseded by 2248/562; **stage-45/46** used the wrong outer-dart owner; and a historical script hash mismatch in the dependency audit was *diagnosed*, not silently repaired.

8.4 The honest reading of the gap

Two numbers frame the situation. The configuration-free best lower bound is 0.11148, which is 1.55% below the engine's accepted bound 0.113235; and the accepted interval is [0.113235, 0.1154469548855829], a relative width of about 1.92%. The engine is therefore much stronger than any single analytic inequality available, and the remaining work is not a matter of more precision or more boxes: it is one inequality for one branch, at one radius, plus — for the exact value — an enumeration and an elimination backend that do not yet exist at $n = 100$.

It is equally important to say what the current bound is *not* sensitive to. Adding boxes, raising an intercept, reusing the $B \geq 37$ intercept, or synchronously retuning the slope and penalty within the frozen family all fail for provable reasons (Part V). The bound is also not limited by compute: the $B = 36$ candidate run takes 130 s of reported time, and the largest single cost in the whole

engine is the tail subdivision. What limits it is that the current local pair expression, at the obstruction point, is genuinely larger than what the $B = 36$ area budget can pay.

Appendix. Evidence ledger, vocabulary and exact arithmetic

A Evidence ledger: assertion ↔ artifact

Assertion	Status	Artifact
100-disk configuration with radius ≤ 0.1154469548855829	replayed	`verify/out/external-disk100-bounds-67ebf1c.json`
Strict margin of that configuration over the reference bound	replayed	`verify/out/external-disk100-margin-67ebf1c.json`
Independent quadtree cross-check at $115447/10^6$	replayed	`verify/out/external-disk100-local-quadtree/replay.json` (parent repository)
Isolated KKT root, strict local minimality mod rotation	replayed	`verify/out/external-disk100-strict-local-67ebf1c.json` (parent repository)
Lattice candidate satisfies $\alpha^2 = 1/67$	independent replay	`verify/out/independent_lattice97_upperbound_n100.json`
$B31$ exclusion	PASS	`verify/out/n100-B31-at-11324.json`
$B32$ exclusion ($M \leq 25$ and $M \geq 26$)	PASS	`verify/out/n100-B32-complete-149.json`
$B33$ exclusion (low M, coherent deep-5, 108 ring models)	PASS	`verify/out/n100-B33-complete-167-v2.json`
$B34$ exclusion (23 chordless + 47 one-chord)	PASS	`verify/out/n100-B34-complete-180.json`
$B35$ exclusion (low M, $M25$, $M26$, $M \geq 27$)	PASS	`verify/out/n100-B35-complete-203.json`
$B \geq 36$ at $R_\star$	PASS	`verify/out/n100-tail-candidate-224.json`
$B \geq 37$ at $R = 2831/25000$	PASS	`verify/out/n100-tail-B37plus-210-a.json`
Candidate scalar budget conservation	PASS	`verify/out/n100-candidate-budget-222.json`
Six branches closed at $R_\star$	PASS	`verify/out/n100-candidate-complete-225.json`
Six-branch ledger with no pending branch	PASS	`verify/out/n100-six-branch-ledger-226.json`
0.11324 proved	**not proved**	`verify/out/n100-tail-support-obstruction-206.json`
0.113236 / 0.113237 proved	**not proved**	`docs/result-n100-stage229-current-best.md`
Integer-edge-count bound $554570693/5000000000$	accepted (analytic + arithmetic)	`verify/out/strengthened_lowerbound_n100.json`
Sharpened-envelope bound $1111720153/10^{10}$	accepted	`verify/out/n100-algebraic-bounds-bundle.json`

Assertion	Status	Artifact
Long-chord bound 2787/25000	accepted (top-level bundle; its geometry sub-block records the numeric threshold as reported, not recomputed)	`verify/out/n100-longchord-bundle-frozen.json`
Area bound 1/10	accepted	`verify/out/lowerbound_n100.json`
Continuous coverage at $R = 153/1250$	`PASS`	`verify/out/coverage_n100.json`
Independent continuous-coverage replay	verified	`verify/out/independent_coverage_n100.json`
Coverage control battery	12/12 passed	`verify/out/coverage_controls_n100.json`
Angular-energy route unavailable	counterexample	`verify/out/n100-angular-energy-obstruction-audit.json`
Global Dirichlet energy transfer fails	`H0_PSD = false`	`runtime/n100-vonequinox-transfer/certificate.json`
Uniform witness dual cannot reach > 100	refuted	`verify/out/n100-witness-dual-obstruction.json`
Boundary-ownership connectivity pruning	refuted (counterexample)	`verify/out/n100-disconnected-boundary-owner-counterexample.json`

B Certificate vocabulary

Field	Meaning
`status`	`PASS`, `PASS_PREMISES`, `NOT_CLOSED`, `PASS_OBSTRUCTION_TO_OLD_RELAXATION`, `PASS_REDUCTION`
`scope`	the exact domain in which the status holds; a `PASS` asserts nothing outside it
`radius`	the candidate radius at which the certificate was established
`required_center_norm_squared_upper`	outward rational enclosure of $1 - R^2$
`minimum_area_margin` / `minimum_reported_positive_margin`	the smallest exact rational margin in the branch
`physical_edge_pairs`	number of shared physical edges whose opposite half-edge terms cancel
`hinge_payment`, `variance_payment`	whether the branch spends its one hinge or variance budget
`geometric_dependencies`	the prose-level analytic lemmas the branch inherits
`machine_verified_geometry`	`false` in every certificate here: finite arithmetic is machine-checked, the geometric reduction is not
`global_optimality`, `global_bound_improved`	`false` in every branch certificate; `true` only in the assembly certificates
`bindings`	source/input SHA-256 map
`not_claimed`	an explicit list of what the artifact does not assert

C The $B = 36$ obstruction and the scalar budget, exactly

C.1 The obstruction

At $R = 2831/25000$ with $\lambda = 3/50$, $\mu = 92531/10^7$, marked constant $K = 973/3125$ and old intercept $L = 1501/200000$, the point $x = y = 9921033/12500000$ satisfies

$$\frac{3754996055077}{500000000000000} \;\le\; \text{pair} \;\le\; \frac{3754996055079}{500000000000000},$$

so

$$\text{pair} - L \;\ge\; \frac{2496055077}{500000000000000} \approx 4.99211 \times 10^{-6},$$

and the $B = 36$ area budget permits at most

$$L_{\max} = \frac{270199092634633283031824697789679316059404571205742262 19}{360000000000000000000000000000000000000000000000000000000000000000},$$

giving an excess of

$$\frac{160623330910716968175302210320683940595428794257737 81}{360000000000000000000000000000000000000000000000000000000000000000} > 0.$$

The direct-term decomposition at the same point is

$$\lambda\text{-distance} \in \Big[\frac{1666193587123}{10^{15}}, \frac{416548396781}{250000000000000}\Big], \quad \tfrac{1}{2}H \in \Big[\frac{3338279809719}{500000000000000}, \frac{83456995243}{12500000000000}\Big],$$

$$\mu\text{-angle penalty} \in \Big[\frac{416380548203}{500000000000000}, \frac{832761096407}{10^{15}}\Big], \qquad \frac{\text{distance}}{R^2} = \frac{27769893118719}{12823297600000}.$$

C.2 The budget as implemented, and an independent recheck

The scalar budget used by the candidate scripts is

$$\text{margin} = (1-\mu)\pi_{\text{lo}} - \big((2m-2-B)A + BK\big)R^2 - BL, \qquad \text{gain} = (A-K)R^2 - L,$$

with $m = 100$ and $A = 129903810568/10^{11}$. Substituting the frozen constants:

Candidate	R	L	margin / gain (decimal)
stage 221/222	$113237/10^6$	$75097/10^7$	$4.906445222464608 \times 10^{-7}$ / $5.1549192230158705 \times 10^{-3}$
candidate 224	22647/200000	939/125000	$1.8087101250224213 \times 10^{-5}$ / $5.152171860143971 \times 10^{-3}$

Both reproduce the stored certificate values exactly, which is how the formula was checked. The stored exact margins are

$$\text{candidate 222:} \quad \frac{490644522246460784697789679316059404571205742262 19}{10^{56}},$$

$$\text{candidate 224:} \quad \frac{180871012502242118246977896793160594045712057422621 9}{10^{56}},$$

and the stored per-cell gains are $64436490287698376249/12500000000000000000000$ and $2576085930071985489/500000000000000000000$.

D Consolidated branch margins

Margins are listed per radius; they are computed at that radius and must not be compared across radii or added.

- $B31$: at $R = 1132301/10^7$, $\frac{3791306959813284956819}{10^{24}}$; at $R = 2831/25000$, $\frac{3205113070935301276991}{10^{24}}$; at $R_\star$ inherited.
- $B32$: at $R = 1132301/10^7$, $\frac{40052428178643350631}{125000000000000000000000}$; at $R = 2831/25000$ positive, with the 73 original rows at least 0.00007800 and the 8 new rows more than 0.00012768; at $R_\star$ inherited.
- $B33$: at $R = 1132301/10^7$, $\frac{141216899554184842489}{10^{24}}$; at $R = 2831/25000$ at least $\frac{1471448836581788939}{500000000000000000000000}$; at $R_\star$ inherited.
- $B34$: at $R = 1132301/10^7$, $\frac{57252253086154121097}{250000000000000000000000}$; at $R = 2831/25000$ at least $\frac{190482881108285619519923}{292968750000000000000000000}$; at $R_\star$ inherited.
- $B35$: at $R = 1132301/10^7$, $\frac{37566017111533099749}{500000000000000000000000}$; at $R = 2831/25000$ at least $\frac{16969003208288867265186533393 1}{625000000000000000000000000000000}$; at $R_\star$ inherited.
- $B \geq 36$: at $R = 1132301/10^7$, $\frac{336050926141338909263097789679316059404571205742262 19}{10^{56}}$; at $R = 2831/25000$, $B = 36$ is obstructed while $B \geq 37$ gives $\frac{166438291677217339982469778967931605940457120574226219}{10^{56}}$; at $R_\star$, $B = 36$ gives $\frac{180871012502242118246977896793160594045712057422621 9}{10^{56}}$.

  The $B31$ margin at $R = 11323/100000$ is $\frac{3797228001573004612011}{10^{24}}$.

E Glossary

Term	Meaning
boundary arc	maximal positive-measure connected segment of ∂D with a.e. constant owner; its count is B
capped	propagation reached its round limit without a fixed point; retained, not decided
capped-pair relaxation	the local pair loss used for the tail branch $B \geq 36$
certificate	a structured machine artifact with a status, a scope, counts and hashes
core	the induced disk remaining after peeling the outer boundary band
deep / ring vertex	a vertex strictly interior to the core, resp. on its boundary
ear	a core face block that can be deleted while preserving the star property
bad face ($q1$)	a marked face of the type counted as a deficit
hinge / variance budget	the single scalar budget a branch may spend once
monotonicity inheritance	using an exclusion proved at a larger radius at a smaller one
obstruction point	an exact rational point where a relaxation exceeds its allowed budget
outward rounding	interval endpoints rounded away from the true value
witness	a point or set of points used to derive a necessary condition

F Audit record for this document

The following checks were executed while preparing this monograph, and their results are reported in the text rather than only in this appendix:

1. `python reproduce.py` — exit 0; accepted bound 0.113235; 0.11324 retained as not closed; four certificate hashes printed (Part VII, the entry-point subsection).
2. **Artifact-coverage audit** — of 100 paths referenced by the mirror's own documents, 19 exist in the mirror and 81 do not; 80 of the 81 exist in the parent repository; `n100-public/proofs/` is absent while the referenced proof exists in the parent repository with a SHA-256 matching the certificate's `proof_sha256` (Part VII, the artifact-coverage subsection).
3. **Binding audit** — the independent lattice receipt binds a stale input and proof (`7e732e18...` and `a0454013...`) while its checker binding (`9655d038...`) matches the current script (Part VII, the binding-audit subsection).
4. **Numeric re-derivation** — the scalar budget formula was re-evaluated in exact rational arithmetic at both candidate radii and reproduces the stored margins and gains exactly (Appendix C).
5. **Fraction-digit audit** — the $B = 36$ margins are numerators of 52 digits over 10^{56}; the obstruction's area-allowed intercept is a 56-digit numerator over $36 \cdot 10^{55}$; the covering squared-radius bound is a 31-digit numerator over 10^{33} equal to $U_\star^2$.
6. **Transplant-package check** — the four sources cited by the branch certificates are present in the parent repository's transplant package and absent from the mirror.

Remark 10. This appendix deliberately lists the failed and retracted items alongside the accepted ones. A record that only ever confirms itself has not been tested; the value of the list above is that it shows where the programme's own machinery was wrong and what replaced it.